



Draft Aquatic Biodiversity and Delineation Report

Proposed construction of Walmer Housing
Development on Erf 11305, Port Elizabeth, in the
Eastern Cape Province

Prepared for:

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DECLARATION OF INDEPENDENCE

I, Roy de Kock as duly authorized representative of BlueLeaf Environmental (Pty) Ltd, hereby confirm my independence (as well as that of BlueLeaf) as a specialist and declare that neither I nor BlueLeaf have any interest, be it business, financial, personal or other, in any proposed activity, application or appeal in respect of which Blueleaf was appointed as aquatic specialist in terms of the National Environmental Management Act, 1998 (Act No. 107 of 1998), other than fair remuneration for worked performed, specifically in connection with the proposed construction of Walmer Housing Development on Erf 11305, Port Elizabeth, in the Eastern Cape Province. I further declare that I am confident in the results of the studies undertaken and conclusions drawn because of it – as is described in this report.



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Experience (years/ months): 19 years

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EXPERTISE

Roy has over 19 years' experience in environmental consulting and specialist services in South Africa. Various projects throughout Africa have also been undertaken. Projects include baseline studies, impact assessments and compliance auditing for various large-scale projects including numerous wind farms, roads (National and Provincial), and infrastructure development projects. Blue Leaf also offers a wide range of in-house specialties including but not limited to Ecological and Botanical assessments, Biodiversity studies, Plant and Animal Search and Rescue, Fauna and Flora permits, Aquatic Assessments, Agricultural and Soil Assessments and Environmental and Venomous animals training workshops.

Roy holds a BSc Honours in Geology and an MSc in Botany from Nelson Mandela University in Port Elizabeth. He is currently busy with his PhD (Doctorate degree) in Botany and Soil Science. He has over 19 years' experience in environmental consulting focusing on Ecological and Agricultural Assessments, Geological and Geotechnical analysis, Environmental Management Plans, mining applications and various environmental impact studies.

Roy has worked as an aquatic biodiversity specialist since 2019 and has worked on numerous projects in the Western, Eastern and Northern Cape and Free State, writing specialist assessments. Some of the more recent aquatic biodiversity assessments are listed below. Agricultural related projects are included:

NAME OF PROJECT	DESCRIPTION	DATE COMPLETED	DETAILS OF CLIENT
Kareedouw Housing Development	Development of new housing areas in Kareedouw, EC	January 2026	Digital Soils Africa Ms Natalie Sharp – 082041401472; natalie@dsafrica.co.za
Woodlands Housing Development	Development of new housing areas in Woodlands, EC	January 2026	

NAME OF PROJECT		DESCRIPTION	DATE COMPLETED	DETAILS OF CLIENT
Louterwater Development	Housing	Development of new housing areas in Louterwater, EC	October 2025	
Joubertina Development	Housing	Development of new housing areas in Woodlands, EC	October 2025	
Mizpah Development	Piggery	Development of a new commercial piggery on a farm outside Humansdorp, EC	January 2026	Habitat Link Roberto Almanza – 082 930 8711; roberto@habitatlink.co.za
Holm Hill Cemetery		Development of a new private cemetery in East London	September 2025	Ayampa Sustainable Development Ayanda Mpambani – 084 456 7249; ayampaprojects@gmail.com
Carolina Smallholdings and Housing Development		Subdivision of a farm into smallholdings and housing near Keiskamma Mouth	May 2025	Abantu Environmental Sive Mlamla – 078 207 8278; info@abantuenvironmental.co.za
SunRiver Development	Citrus	New citrus expansion on farm SunRiver near Kirkwood, EC	May 2025	Habitat Link Roberto Almanza – 082 930 8711; roberto@habitatlink.co.za
Kou-Kamma Two Rivers Piggery		Development of a new commercial piggery in Joubertina, EC	March 2025	Habitat Link Roberto Almanza – 082 930 8711; roberto@habitatlink.co.za
Nxele Piggery		Development of a new commercial piggery in Uitenhage, EC	April 2025	
Prashanti Poultry		Development of a new commercial poultry facility in Rocklands, EC	April 2025	
Sam Piggery		Development of a new commercial piggery facility in Rocklands, EC	April 2025	
Schaapkraal Poultry		Converting a vegetable farm into poultry in the Phillipi area, WC	March 2025	Rafuni (Pty) Ltd Nyakallo – 079 178 8701; geoenviro@rafuni.co.za
Thembaletu Agriculture	Mix	Formalization of 346 ha farmland to farm vegetables, maize, legumes and livestock near George, WC	January 2025	
Zolani Small Scale Farming		Development of mixed farming in piggery, sheep, goats, chickens etc on various farms in Ashton, WC	March 2025	

This study complies with the requirements listed in the Gazetted protocols for the specialist assessment (GN. R 320 of 2020) and minimum report content requirements for environmental impacts on Aquatic Biodiversity.

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DRAFT

1. Introduction

BlueLeaf Environmental (Pty) Ltd has been appointed by Abantu Environmental Services (Pty) Ltd, to conduct an Aquatic Biodiversity and Delineation Report as part of the Environmental Impact Assessment (EIA) process conducted by Abantu. This report also complies with the Department of Water and Sanitation's (DWS) requirements for a wetland delineation and assessment report to confirm the presence of wet conditions and the extent thereof on the proposed development.

This investigation has been undertaken to form part of the National Environmental Management Act (NEMA) (Act No 107 of 1998) as amended, as well as the Integrated Water Use License Application (IWULA) to be submitted to the Department of Water and Sanitation (DWS). The establishment of any type of development on the extent of the rivers, wetland systems and floodplains within the property poses detrimental environmental impacts based on the nature and magnitude of the development and legal requirements in the form of the NEMA, and the National Water Act. (NWA; Act 36 of 1998).

1.1. Project location

The proposed housing development will be located on Erf 11305, Port Elizabeth, in the Nelson Mandela Bay Metropolitan Municipality (Port Elizabeth/Gqeberha) in the Eastern Cape Province (Figure 1.1).

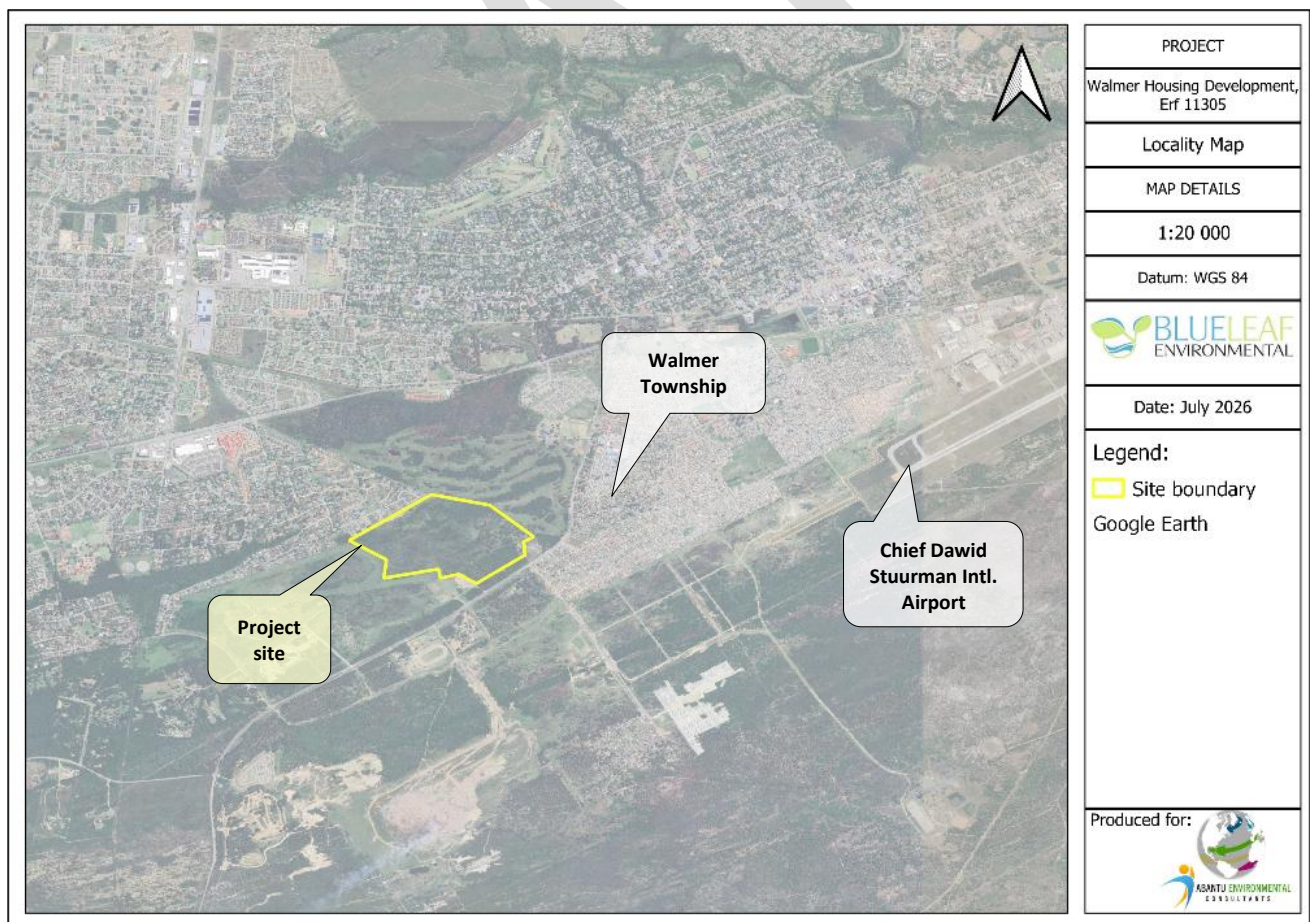


Figure 1.1: Location map of the property for proposed development in St Francis Bay

The development site is located between the suburbs of Walmer Heights and Beaumont Estates, to the west and northwest; the old Arlington Racecourse, to the west; SmartStone Port Elizabeth, to the

south; the Walmer Country Club, to the north and northeast; Walmer Township, to the east, and Victoria Drive (M18) and open space, to the southeast.

1.2. Project description

The project will involve construction of the Walmer Housing Development, including associated facilities and infrastructure. The project site is 43.73 hectares in total extent. See site layout plan shown in Figure 1.2.

The Gross Residential Area (single dwellings) is 11.11 hectares – with a Gross Residential Density of 14.9 units per hectare (single dwellings) and a Nett Residential Density of 58.69 units per hectare (single dwellings). The average erf size for single dwellings is 170.41 m². No internal service infrastructure is currently in place on site – however, electrical, sewage and water infrastructure will be connected to the existing bulk infrastructure currently servicing the surrounding residential areas.

The housing development will provide a mix of housing typologies and associated facilities which include the following:

Housing typologies and associated facilities:

- Free Basic House/RDP
- GAP/FLISP
- Social Housing
- Open/GAP market housing

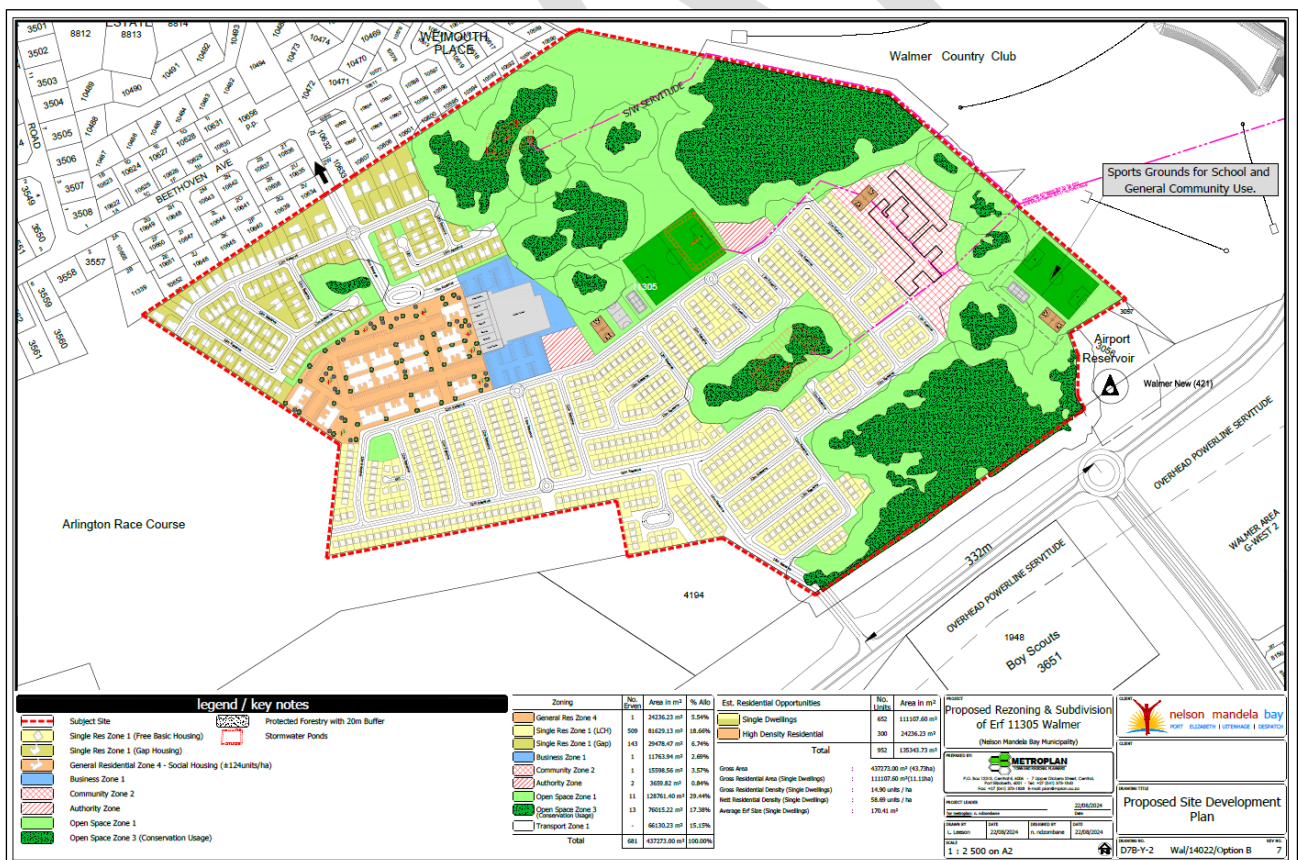


Figure 1.2 Site layout plan for the proposed housing development.

Other land uses:

- Business Zone I: to provide for businesses such as a supermarket, bottle store, service trades etc. with parking provision at 3.33 bays per 100 m²
- Institutional Zone I: school site
- Authority Zone: authority uses such as police station, fire services, telecommunications facilities etc.
- Open Space Zone I: public open space.
- Special Use: conservation usage (forest patches)
- Transportation Zone II: public zone.

The total area of the site is 43.73 ha. The Gross Residential Area (single dwellings) is 11.11 ha with Gross Residential Density (single dwellings) being 14.90 units/ha and a Nett Residential Density (single dwellings) being 58.69 units/ha. The average erf size for single dwellings is 170.41m². No internal infrastructure is currently in place on site, however electricity, sewer and water will be connected onto existing bulk infrastructure currently servicing the surrounding areas.

1.3. Legislative context

The following legislation is directly relevant when assessing the aquatic environment relating to the proposed development.

National Web based Environmental Screening Tool

The National Web based Environmental Screening Tool (<https://screening.environment.gov.za/screeningtool/>) is a geographically based web-enabled application which allows a proponent intending to apply for environmental authorization in terms of the Environmental Impact Assessment (EIA) Regulations 2014 (as amended), to screen their proposed site for any environmental sensitivity.

The Screening Tool also provides site specific EIA process and review information, for example, the Screening Tool may identify if an industrial development zone, minimum information requirement, Environmental Management Framework or bio-regional plan applies to a specific area.

Further to this, the Screening Tool identifies related exclusions and/ or specific requirements including specialist studies applicable to the proposed site and/or development, based on the national sector classification and the environmental sensitivity of the site.

Finally, the Screening Tool allows for the generating of a Screening Report referred to in Regulation 16(1)(v) of the Environmental Impact Assessment Regulations 2014, as amended whereby a Screening Report is required to accompany any application for Environmental Authorization and as such the tool has been developed in a manner that is user friendly and no specific software or specialized GIS skills are required to operate this system.

Procedures for the assessment and minimum criteria for reporting on identified environmental themes in terms of sections 24(5)(a) and (h) and 44 of the National Environmental Management Act, 1998, when applying for environmental authorization have been Gazetted (GN. R 320 of 20 March 2020). In terms of sections 24(5)(a), (h) and 44 of the National Environmental Management Act, 1998. These procedures prescribe general requirements for undertaking site sensitivity verification and for protocols for the assessment and minimum report content requirements of environmental impacts for environmental themes for activities requiring environmental authorization, as contained in the Schedule therein. When the requirements of a protocol apply, the requirements of Appendix 6 of the

Environmental Impact Assessment Regulations, as amended, (EIA Regulations), promulgated under sections 24(5) and 44 of the National Environmental Management Act, 1998 (Act No. 107 of 1998), are replaced by these requirements.

The contents of this specialist report comply with the legislated requirements as described in the following environmental theme and associated specialist assessment protocol as listed in the project's Screening Tool Report:

➤ ***Protocol for the Specialist Assessment and Minimum Report Content Requirements for Environmental Impacts on Aquatic Biodiversity.***

National Environmental Management Act (NEMA) (107 of 1998; as amended), and the EIA regulations (as amended):

Although the Specialist Assessment Protocols (as listed above) supersedes this legislative requirement, the contents of this specialist report still comply with the legislated requirements as described in Appendix 6 of the National Environmental Management Act (No 107 of 1998; NEMA) Regulations of 2014 and updated in 2017 (GN R. 326 of 2017).

Other national legislation

Articles of legislation applicable to this proposed development include:

Title of legislation or guideline	Administering authority	Applicability to the project
National Environmental Management Act (NEMA) Environmental Impact Assessment (EIA) Regulations 2014 as amended (Act No. 107 of 1998)	DEDEAT	The activity triggers activities listed in NEMA EIA Regulations
National Water Act, 1998 (Act No. 36 of 1998)	Department of Water & Sanitation (DWS)	Infrastructure may impact on existing surface water systems. Requirements for a Water Use License/ General Authorization are required.
Section 21 (i) of the National Water Act: Altering the bed, banks or characteristics of a water course	DWS	This form allows the applicant to provide information about their water use in respect of Physical changes that are made to a water course, for example to widen or straighten the channel of a river. Alteration of the bed and banks is usually needed for construction and infrastructure development near or across a river. Sand mining is another common example of this water use. Alteration of the course of a watercourse refers to the diversion of the water course. The river channel is usually reconstructed or replaced with a canal which may extend several kilometers from the original course.
Section 21 (c) of the National Water Act: Impeding or diverting the flow of water in a watercourse	DWS	Impeding or diverting flow does not cause any loss.

Title of legislation or guideline	Administering authority	Applicability to the project
Government Notice 509 as published in the Government Gazette 40229 of 2016	DEDEAT/DWS	Defines the area of a watercourse as: - the outer edge of the 1 in 100-year flood line and/or delineated riparian habitat, whichever is the greatest distance, measured from the middle of the watercourse of a river, spring, natural channel, lake or dam. - in the absence of a determined 1 in 100-year flood line or riparian area the area within 100 m from the edge of a watercourse where the edge of the watercourse is the first identifiable annual bank fill flood bench: or - a 500m radius from the delineated boundary (extent) of any wetland or pan in terms of this regulation.

Relevant Provincial and Metropolitan legislation include:

Title of legislation or guideline	Administering authority	Applicability to the project
The Eastern Cape Biodiversity Conservation Plan (ECBCP) – Freshwater map	South African National Biodiversity Institute (SANBI)	Listing of freshwater CBAs and ESAs within the study site.
The National Freshwater Ecosystem Priority Areas (NFEPA) Project	South African National Biodiversity Institute (SANBI)	Identification and classification of any sensitive river or wetland system within 500 m of the study site boundary.

1.4. Alternatives

No site or layout alternatives are proposed.

1.5. Public consultation

No consultation requirements were identified during the drafting of this specialist report. The findings of this report can be presented to stakeholders and I&APs as part of the BAR Public Participation Process (PPP).

No comments were received to date on this report.

1.6. Objectives

The objectives of the project are listed below. These objectives are based on the requirements of each specialist protocol as listed in the DFFE Screening Report:

- Describe both the existing area as well as the area prior to construction in terms of its current ecological characteristics and the general sensitivity of these components to change.
- Confirm if there are any outright fatal flaws to the establishment of the proposal at its current location from a biodiversity perspective.
- Map all existing areas to be directly affected by the proposals in terms of its current and previous biodiversity sensitivity (constraints).

- Map all 'No-Go' areas.
- Describe the likely scope, scale, and significance of impacts (positive and negative) on biodiversity components of the area associated with the construction of the proposals.
- Make recommendations on the scope of any mitigation measures that may be applied during construction to avoid/reduce the significance of the identified construction-related impacts.
- Describe the likely scope, scale, and significance of impacts (positive or negative) on the biodiversity components associated with the operation or use of the proposals.
- Make recommendations on the scope of any mitigation measures that may be applied to avoid/reduce the significance of the operations-related impacts. These mitigation measures could also be design recommendations as well as operational controls, monitoring programmes, management procedures and the like.
- It will be particularly important to identify any rehabilitation measures that can be reasonably applied on the completion of the construction works.
- Broadly comment on the cumulative ecological impacts (positive or negative) associated with the construction and/or operation of the proposals.

It should be noted that only datasets and base data relevant to the study area and affected environmental features are discussed below.

1.7. Assumptions and limitations

The processes of investigation which have led to the production of this report, harbours several assumptions, which include the following:

- The report is based on currently available information and, as a result, limited by the information provided by the Client.
- The report is limited by seasonality as the presented data will be based on a single site survey of the site characteristics conducted within a single season (early spring) of a single year (2024).
- A desktop delineation of the wetland and riparian area was done before the site visit. This is thought to be an acceptable method.
- The watercourses were mainly dry and aquatic assessments such as water quality measurements, macroinvertebrate (South African Scoring System Version 5 -SASS5), Integrated Habitat Assessment System (IHAS), Macro-Invertebrate Response Assessment Index (MIRAI version 2) and fish sampling (Fish Response Assessment Index - FRAI) were excluded.
- The watercourse assessment is confined to the proposed project footprint and does not include the neighbouring and adjacent properties, which were only considered as part of the desktop assessment.
- Global Positioning System (GPS) technology is inherently inaccurate and some inaccuracies due to the use of handheld GPS instrumentation (Garmin eTrex) may occur. If more accurate assessments are required, the watercourses will need to be surveyed and pegged according to surveying principles.
- The risk assessment was applied on the basis that the stipulated mitigation measures and all specialist recommendations will be implemented, and therefore the results presented demonstrate the impact significance of perceived impacts on the receiving freshwater environment post-mitigation.
- All information provided by the EAP, applicant and engineering design team to the environmental specialist was correct and valid at the time that it was provided.
- The proposed project footprint as provided by the engineering design team is correct and will not be significantly deviated from.

- Significant reliance on visual eco-morphological observations was made to derive an understanding of the state of the habitat within the subject site. This state may change under a different meteorological regime.
- Freshwater resources that fall outside of the affected catchment (but still within the 500 m DWS regulated area) and are not at risk of being impacted (such as upslope water resources) by the specific activity were not delineated nor assessed. Such features were flagged during a baseline desktop assessment prior to the site visit.
- This assessment deals primarily with inland wetlands (i.e. no existing connection to the ocean and these ecosystems are characterised by the complete absence of marine exchange and/or tidal influence) (Ollis et al., 2013).
- Selection of assessment techniques and tools were based on the assessment practitioner's knowledge and experience of these tools and their attributes and shortcomings.
- The assessment techniques and tools are currently the most appropriate available tools and techniques to undertake assessments of freshwater resources; they are rapid assessment tools that rely on qualitative information and expert judgment. While these tools have been subjected to peer review processes, the methodology for these tools are ever evolving and will likely be further refined in the future. For the purposes of this assessment, the assessments were undertaken at rapid levels with somewhat limited field verification: it therefore provides an indication of the PES of the portions of the affected systems rather than providing a definitive measure.
- PES and EIS were only determined for the affected/regulated areas even though upstream and downstream as well as catchment impacts were considered (based on available desktop information).
- The PES and EIS assessments undertaken are largely qualitative assessment tools and thus the results are open to professional opinion and interpretation.
- The EIS assessment did not specifically address the finer-scale biological aspects of the watercourses such as faunal species.
- The initial study was undertaken as a desktop assessment and as such, the information gathered must be considered with caution, as inaccuracies and data capturing errors are often present within these databases.
- The guideline document, "A Practical Field Procedure for the Identification and Delineation of Wetlands and Riparian Areas" document, as published by DWAF (2005) was followed for the delineation of the wetland and riparian areas.
- This assessment is focussed on surface water and excluded terrestrial landscapes and groundwater features.
- BlueLeaf Environmental is an independent environmental consulting firm and as such, all processes and attributes of the EIA are addressed in a fair and unbiased fashion. It is believed that through the running of a transparent and participatory process, risk associated with assumptions, uncertainties and gaps in knowledge can be, and were, minimised.

1.8. Project Area of Influence

The Project Area of Influence (PAOI) is defined according to important ecosystem processes and functions that may be plausibly affected by the proposed development and its associated activities. The PAOI sets the minimum spatial extent of the study area, and the assessment will be focused within this area. The following site descriptors were used to delineate each PAOI (Figure 1.3).

PAOI	Area (ha)	Description	Probability of impact occurring
Primary PAOI	43.6 ha	The Primary PAOI includes all areas within the boundaries of the project development site. This is the area directly impacted by the proposed development and includes both permanent and temporary footprints.	Definite
Secondary PAOI	225.4 ha	The secondary PAOI includes all areas within a 500 m buffer of the proposed development boundary. These areas are not directly impacted by the development unless temporary footprints like site camps, laydown areas and stockpiles are placed in them. Assessing this PAOI will not only result in identifying potential indirect and cumulative impacts but will also allow for micro-movement of infrastructure.	Unlikely
Total PAOI	269 ha	The Primary and Secondary PAOI's are collectively referred to as the Total PAOI (or just the PAOI) or Study Site in this report and demarcate the extent of the study site that will be assessed.	Likely

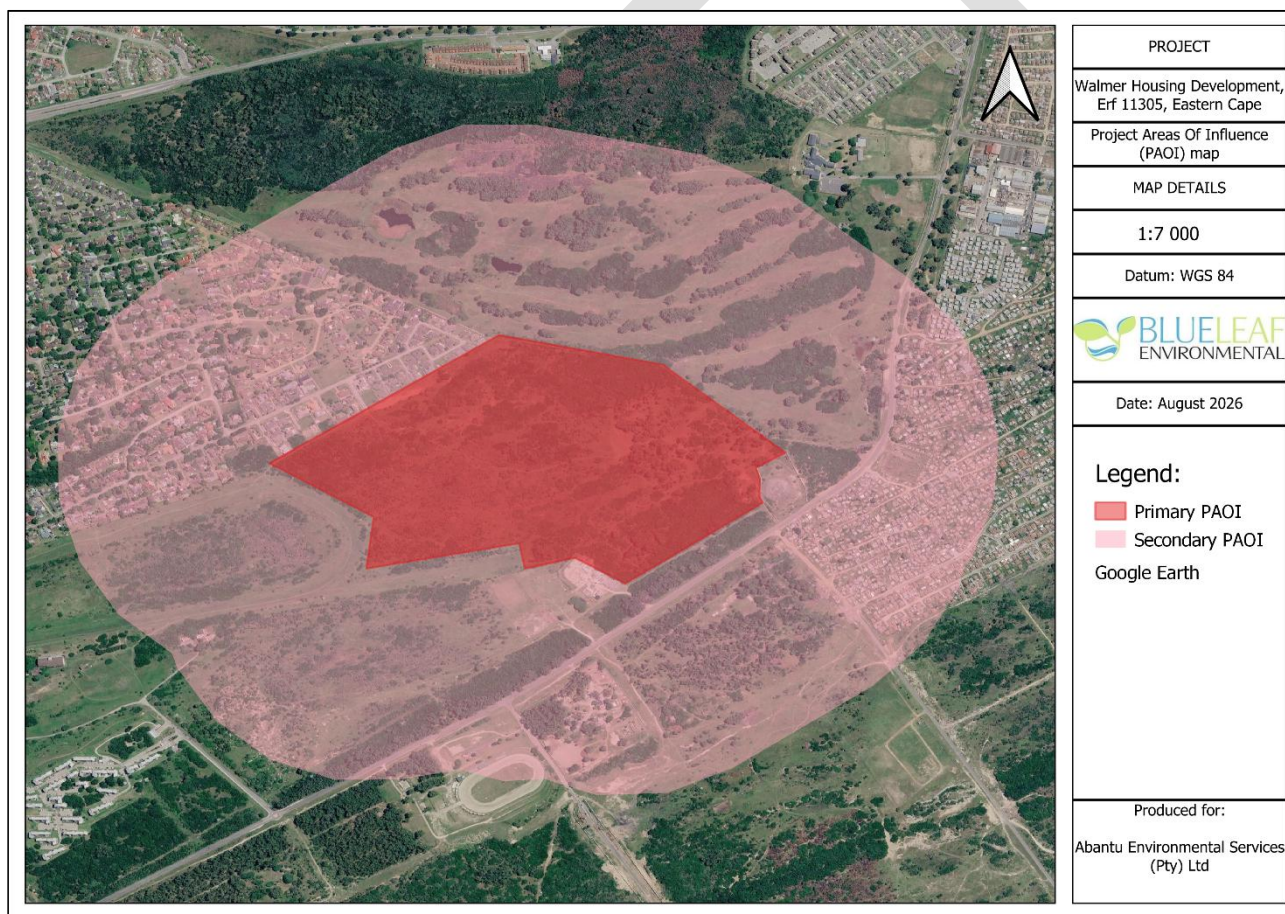


Figure 1.3: Site map showing the PAOI of the project site and surrounding areas.

2. Approach and Methodology

The aim of this assessment is to identify areas of aquatic importance and to evaluate these in terms of their conservation importance. To do so, the aquatic sensitivity of the area is assessed. The condition and sensitivity of the vegetation will also largely determine areas with high biodiversity.

The study site and surrounding areas were assessed using a two-phased approach. Firstly, a desktop assessment of the site was conducted in terms of current biodiversity programs and plans (listed further below).

Further to the above, a site visit was conducted on the 27th of July 2026. The site visit served to inform potential impacts associated with the proposed project and how significantly it would impact on the surrounding environment.

The aim of this study is to identify areas of high sensitivity and those that may be subject to significant impacts from the project. Aspects that would increase impact significance include:

- Riparian vegetation
- The presence of freshwater process areas.
- Aquatic corridors.
- Aquatic classification

2.1. Vegetation mapping

Mucina and Rutherford (2006) developed the National Vegetation Map (VegMap). The latest update of the VegMap took place in 2024. This map describes each vegetation type in detail, along with the most important species including endemic species and those that are biogeographically important. This is the most comprehensive data for vegetation types in South Africa. The conservation status of ecosystems was further extracted from the 2018 National Biodiversity Assessment Synthesis Report (South African National Biodiversity Institute (SANBI; 2019) and the National List of Ecosystems that are Threatened and in Need of Protection (GN 1002 of 9 December 2012).

2.2. Classification of Freshwater systems

The following datasets were used to identify, delineate, and classify any surface water bodies found within the study site or within 500 m of the study site:

Table 2.1: Legislation relevant to the identification, delineation, and classification of surface water bodies

Relevant legislation	Discussion
Department of Water Affairs and Forestry (DWAF) (2008): A practical Guideline Procedure for the Identification and Delineation of Wetlands and Riparian Zones.	Describing and delineating all watercourses with the study site.
GN 509 as published in the Government Gazette 40229 of 2016 as it relates to activities as stipulated in Section 21(c) and (i) of the National Water Act, 1998 (Act No. 36 of 1998)	Delineation of watercourses and wetlands within 500m of the study area.
Classification System for Wetlands and other Aquatic Ecosystems in South Africa. User Manual: Inland systems (Ollis et al., 2013)	Classification of all identified watercourses
Technical Report for the National Freshwater Ecosystem	Identification of FEPAs freshwater ecosystems.

Relevant legislation	Discussion
Priority Areas project (Nel et al., 2011)	

Wetland Assessments

The following definitions, as per the National Water Act, 1998 (Act No. 36 of 1998) are of relevance to this study:

Watercourse means:

- (a) A river or spring.
- (b) A natural channel in which water flows regularly or intermittently.
- (c) A wetland, lake or dam into which, or from which water flows; and
- (d) Any collection of water, which the Minister may, by notice of the Gazette, declare a watercourse.

Wetland means:

“Land which is transitional between terrestrial and aquatic systems where the water table is usually at or near the surface, or the land is periodically covered with shallow water, and which land in normal circumstances supports or would support vegetation typically adapted to life in saturated soil.”

Riparian habitat includes:

“The physical structure and associated vegetation of areas associated with a watercourse which are commonly characterized by alluvial soil, and which are inundated or flooded to an extent and with a frequency sufficient to support vegetation of species with a composition and physical structure distinct from those of adjacent land areas”.

There are many different types of wetlands; and several classification systems have been developed to try to describe these different types. The system used in this report has been developed for inland wetland systems, and is based on the hydrogeomorphic (HGM) characteristics of wetlands (Marneweck and Batchelor, 2002; Kotze et al., 2005)

This approach follows that used by the US Environmental Protection Agency, and this classification system has been included as part of a proposed wetland classification system for South African wetlands by Ewart-Smith et al (2006).

Wetlands can be classified by methods that range from the use of commonly recognized vegetation or cover types to systems based on hydrology, geomorphology, or some combination of the two. The classification presented here is based on the hydrogeomorphic functions of wetlands.

There are three basic properties that are used to provide insight into wetland functions. This hydrogeomorphic classification system classifies wetlands according to their form (geomorph-characteristics) and the way in which water moves in, through and out of the wetland system (hydro-characteristics). The classification system recognises 5 generic palustrine wetland types:

- Pans and depressions (incl. lakes).
- Seepage wetlands.
- Un-channeled valley bottoms.
- Channeled valley bottoms; and
- Floodplains.

Different wetlands perform different functions in the landscape.

Wetland delineation

The DWS (2005) guidelines for “a practical field procedure for delineation of wetlands and riparian areas” are recommended in Gazette No. 19182, Notice No. 1091 of the National Water Act, 1998. This guideline explains the field indicators and methods for determining whether an area is a wetland or a riparian area, and how to find its boundaries. Although the primary driver of a wetland is water, due to its dynamic nature water is not a very useful parameter for identifying the outer boundary of a wetland. What is needed is a method of identifying the indirect indicators of prolonged saturation by water. This includes wetland plants (hydrophytes) and wetland (hydromorphic) soils. Their presence or absence implies the frequency and duration of saturation and is a satisfactory indicator to classify the area as a wetland (DWAF, 2005).

In wetland delineation there are three zones which are distinguished according to a changing frequency of saturation. These are the permanent, seasonal and temporary zones. The primary objective of wetland delineation is usually to define the outer edge of the temporary zone as it marks the boundary between the wetland and the adjacent terrestrial zone. There are four important indicators that are used to define the boundaries of a wetland. The most important one is the soil wetness indicator with terrain unit, soil form and vegetation acting as confirmation. The point where wetland indicators are not present is regarded as the edge of the wetland.

The permanently wet zone is characterized by dark grey, clay soil, caused by a lack of oxygen required for the oxidation of minerals such as iron in the soil. The seasonally wet zone is characterized by grey soils with lots of orange and black mottles. It is generally recommended that there should be a 100m buffer zone between the edge of the delineated temporary zone and any development. Important indicators of each zone are as follows:

1. Wetland vegetation

To tolerate the anaerobic conditions of seasonal or permanent flooding, hydrophytes (water loving plants) have evolved several adaptations. Their presence can therefore indicate a moist soil habitat and thus provide a potential boundary of a wetland's seasonally flooded or permanent flooded zones (Macfarlane *et al.*, 2007).

- The **temporary zone** of a wetland will show mainly grass, some woody species and some sedges.
- The **seasonal zone** will begin to show more hydrophytic (or water loving) sedges with tall grass (over 1m).
- The **permanent zone** will be noticeable by emergent reeds and sedges, bulrushes or floating and submerged plants. Woody species will have adaptations for permanent wetness such as prop roots (Mangroves).

2. Wetland Soil

Low oxygen levels result in a reduced rate of organic matter decomposition within the soil, where Sulphur tends to exist in its reduced form, hydrogen sulphide (H₂S), noticeable by its tell-tale rotten-egg smell. These conditions also serve as a catalyst for the metals in the soil to become soluble and begin leaching (DWAF, 2005). The metals produce rich colors of yellow, orange and reds.

- The **temporary or seasonal zone** of a wetland, where there is more seasonal flooding, produces mottling of colors, as the metals are still in the process of precipitating. These mottles occur within a grey matrix where the metals have already leached.

- The **permanent zone** of a wetland, where there is more permanent flooding of the soil, produces leaching of metals, with soils remaining a grey ("gleyed") color.
- It is recommended by DWS that soils be sampled on the surface (0-10cm) and between 40 and 50cm.

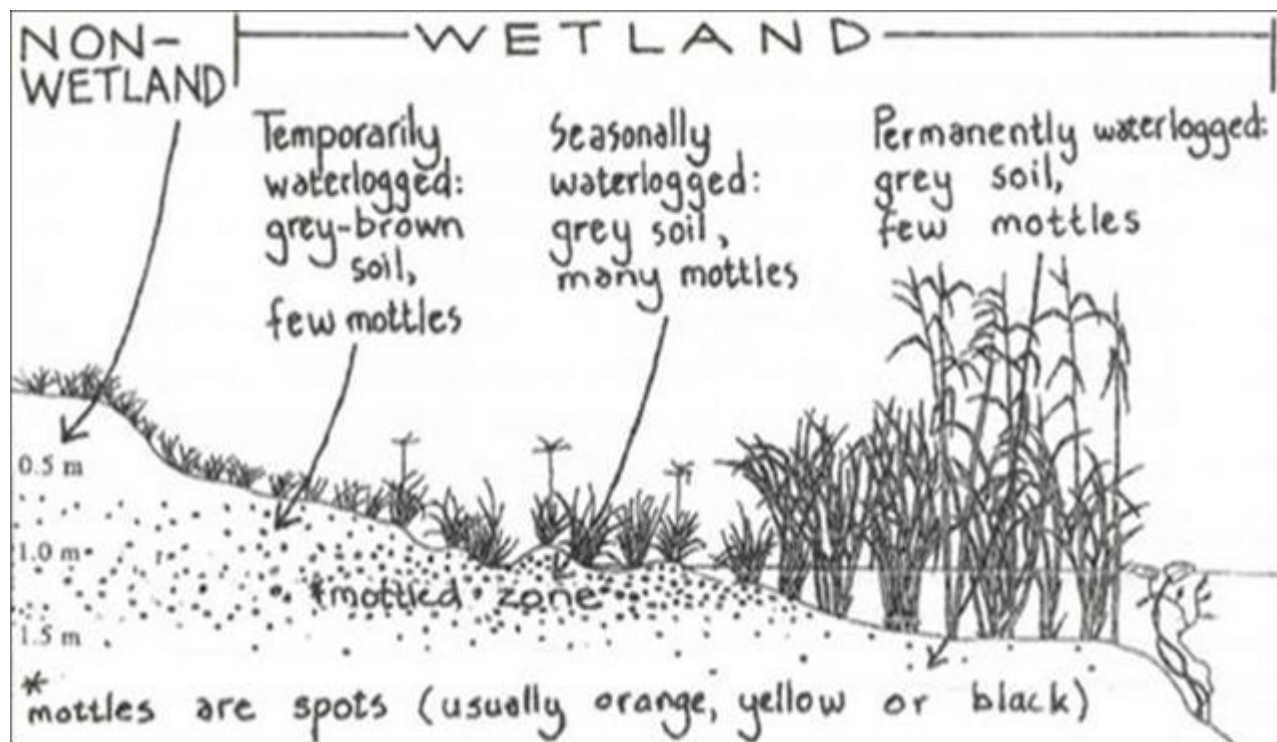


Figure 2.1: A cross-section through a wetland, indicating how the soil wetness and vegetation indicators change as one moves along a gradient of decreasing wetness, from the middle to the edge of the wetland (DWS, 2005).

Present Ecological Status

The Present Ecological State (PES) refers to the current state or condition of a watercourse in terms of all its bio-physical characteristics and reflects the change to the watercourse from its reference condition. The method used to determine the PES for watercourses was the Index of Habitat Integrity (IHI) which measures the impact of human disturbance on riparian and instream habitats (Kleynhans, 1996). The IHI is a rapid assessment of the severity of impacts affecting habitat integrity within a river reach. It can be applied to both perennial and non-perennial watercourses (Dabrowski, 2019; Kleynhans, 1996). Each impact on the riparian and instream habitat is given a score between 0 – 20 based on the degree of modification (Table 2.2 and 2.3).

The results are then compared to the standard DWS A-F ecological categories (Table 2.2) from where the PES/Habitat integrity of the wetland can be determined. The values give an indication of the alterations that have occurred in the wetland system.

Table 2.2: Present Ecological Status categories (adapted from Kleynhans, 1996 and 1999)

Ecological Category	Score	Description
A	90-100%	Unmodified, natural
B	80-90%	Largely natural with a few modifications. A small change in natural habitats and biota may have taken place but the ecosystem functions are essentially unchanged.

Ecological Category	Score	Description
C	60-80%	Moderately modified. Loss and change of natural habitat and biota have occurred, but the basic ecosystem functions are still predominantly unchanged.
D	40-60%	Largely modified. A large loss of natural habitat, biota and basic ecosystem functions is extensive.
E	20-40%	Seriously modified. The loss of natural habitat, biota and basic ecosystem functions is extensive.
F	20-40%	Critically/ Extremely modified. Modifications have reached a critical level, and the system has been modified completely with an almost complete loss of natural habitat and biota. In the worst instances the basic ecosystem functions have been destroyed, and the changes are irreversible.

The method used to determine the PES of wetlands is based on a Rapid Assessment version of WET-Health by Kotze et al., (2009). This method is based on assigning a score between 0-10 to simple criteria (Table 2.2). A PES class is then determined based on the resulting score (Table 2.3).

Table 2.3: Criteria used to determine the PES for wetlands.

PES assessment per driver		
Hydrology	Description	Impact score rating (0-10)
None	No discernible modifications, or the modifications are of such nature that they have no impact on the hydrological integrity	0-0.9
Small	Although identifiable, the impact of the modifications on the hydrological integrity is small	1-1.9
Moderate	The impact of the modifications on the hydrological integrity is clearly identifiable, but limited	2-3.9
Large	The impact of the modifications is clearly detrimental to hydrological integrity. Approximately 50% of hydrological integrity has been lost	4-5.9
Serious	Modifications clearly have an adverse effect on hydrological integrity. 51% to 79% of hydrological integrity has been lost.	6-7.9
Critical	Modifications are so great that the hydrological functioning has been drastically altered. 80% or more of hydrological integrity has been lost	8-10
	Impact score rating (0-10)	
Geomorphology	Description	Impact score rating (0-10)
None	Unmodified, natural.	0-0.9
Small	Largely natural. A slight change in geomorphic processes is discernable, but the system remains largely intact	1-1.9
Moderate	Moderately modified. A moderate change in geomorphic processes has taken place but the system remains predominantly intact	2-3.9
Large	Largely modified. A large change in geomorphic processes has occurred and the system is appreciably altered	4-5.9
Serious	Greatly modified. The change in geomorphic processes is great, but some features are still recognizable	6-7.9
Critical	Modifications have reached a critical level as geomorphic processes have been modified completely	8-10
	Impact score rating (0-10)	
Vegetation	Description	Impact score rating (0-10)
None	Vegetation composition appears entirely natural.	0-0.9
Small	A very minor change to vegetation composition is evident at the site (e.g. abundance of ruderal, indigenous invasive slightly higher than would be the case naturally).	1-1.9
Moderate	Vegetation composition has been moderately altered but introduced, alien and/ or increased ruderal species are still clearly less abundant than characteristic indigenous wetland species	2-3.9
Large	Vegetation composition has been largely altered and introduced, alien and/ or increased ruderal species occur in approximately equal abundance to the characteristic indigenous wetland species	4-5.9
Serious	Vegetation composition has been substantially altered but some characteristic species remain, although the vegetation consists mainly of introduced, alien and/or ruderal species	6-7.9
Critical	Vegetation composition has been almost totally altered, and in the worst case all indigenous vegetation has been lost (e.g. as a result of a parking lot).	8-10
	Impact score rating (0-10)	
	AVERAGE	
Generalized PES assessment		
PES	Description	Impact score rating (0-10)
None	Unmodified, natural	0-0.9
Small	Largely natural with few modifications. A slight change in ecosystem processes is discernable, and a small loss of natural habitats and biota may have taken place.	1-1.9
Moderate	Moderately modified. A moderate change in ecosystem processes and loss of natural habitats has	2-3.9

PES assessment per driver		
	taken place but the natural habitat remains predominantly intact.	
Large	Largely modified. A large change in ecosystem processes and loss of natural habitat and biota has occurred	4-5.9
Serious	The change in ecosystem processes and loss of natural habitat and biota is great but some remaining natural habitat features are still recognizable	6-7.9
Critical	Modifications have reached a critical level, and the ecosystem processes have been modified completely with an almost complete loss of natural habitat and biota.	8-10
		Impact score rating (0-10)

Ecological Importance and Sensitivity

The Ecological Importance and Sensitivity (EIS) of a watercourse is an expression of its importance to the maintenance of ecological diversity and functioning on local and wider scales, and both abiotic and biotic components of the system are taken into consideration.

Sensitivity refers to the system's ability to resist disturbance and its capability to recover from disturbance once it has occurred. The ecological importance and sensitivity categories are indicated in Table 2.5.

For watercourses, the EIS is based on a rapid instream and riparian habitat ecological importance and sensitivity assessment assigning a score between 0-4 to simple criteria (Table 2.4). The level of confidence in the score is given. The average scoring of these criteria places the watercourse in an EIS Category according to Table 2.5.

Table 2.4: Outcome of a rapid instream and riparian habitat ecological importance and sensitivity assessment, using a modified version of DWAF EIS tool for rivers.

Determinant	Score	Confidence	Comments
1. Rare & Endangered Species			
2. Populations of Unique Species			
3. Species/taxon Richness			
4. Diversity of Habitat Types or Features			
5 Migration route/breeding and feeding site for wetland species			
6. Sensitivity to Changes in the Natural Hydrological Regime			
7. Sensitivity to Water Quality Changes			
8. Flood Storage, Energy Dissipation & Particulate/Element Removal			
9. Protected Status			
10. Ecological Integrity			
Total			
Median			
Overall ecological sensitivity & importance			

Table 2.5: Ecological Importance and Sensitivity Categories of Wetlands.

EIS Categories	Score	Description
Low/Marginal	D	Not ecologically important and sensitive at any scale. Biodiversity is ubiquitous and not sensitive to flow and habitat modifications. (Wetlands: play an insignificant role in moderating water quality and quantity).
Moderate	C	Ecologically important and sensitive on provincial/local scale. Biodiversity is not usually sensitive to flow and habitat modifications. (Wetlands: play a small role in moderating water quantity and quality)
High	B	Ecologically important and sensitive and important. Biodiversity may be sensitive to flow and habitat modifications. (Wetlands: play a role in moderating water quantity and quality).
Very High	A	Ecologically important and sensitive on a national (or even international) level. Biodiversity is usually very sensitive to flow and habitat modifications. (Wetlands: play a major role in moderating water quantity and quality.)

The method used to assess the EIS of wetlands includes an assessment of ecological (e.g. presence of rare and endangered fauna/flora), functional (e.g. groundwater storage/recharge) and socio-economic criteria (e.g. human use of the wetland). Scoring these criteria places the wetland in a Wetland Importance Class (A-D) according to Table 2.6.

Table 2.6: Wetland Importance Class integrating Ecological Importance and Sensitivity (EIS), and functional and socio-cultural importance modifiers.

Importance class (one or more attributes may apply)		Range of Median	Wetland Importance Class
Very high Representative of wetlands that: • support key populations of rare or endangered species. • have a high level of habitat and species richness. • have a high degree of taxonomic uniqueness and/or intolerant taxa. • provide unique habitat (e.g. salt marsh or ephemeral pan; physiognomic features, spawning or nursery environments). • is a crucial avifaunal migratory node (e.g. RAMSAR wetlands). • may provide hydraulic buffering and sediment retention for large to major rivers that originate largely outside of urban conurbations. • have groundwater recharge/discharge comprising a major component of the hydrological regime of the wetland. • are highly sensitive to changes in hydrology, patterns of inundation, discharge rates, water quality and/or disturbance; and • are of extreme importance for conservation, research or education.		>3 <=4	A
High Representative of wetlands that: • support populations of rare or endangered species, or fragments of such populations that are present in other similar and geographically adjacent wetlands. • contain areas of habitat and species richness. • contain elements of taxonomic uniqueness and/or intolerant taxa. • contain habitat suitable for specific species (e.g. physiognomic features). • provide unique habitat (e.g. salt marsh or ephemeral pan; spawning or nursery environments, heronries). • may provide hydraulic buffering and sediment retention for rivers that originate largely outside of urban conurbations, or within residential fringes of urban areas. • have groundwater recharge/discharge comprising a component of the hydrological regime of the wetland. • may be sensitive to changes in hydrology, patterns of inundation, discharge rates, water quality and/or human disturbance; and • are important for conservation, research, education or eco-tourism.		> 2 <= 3	B
Moderate Representative of wetlands that: • contain small areas of habitat and species richness. • provide limited elements of habitat that has become fragmented by development (e.g. salt marsh, ephemeral pan; roosting sites and heronries). • provide hydraulic buffering for rivers that originate in urban areas. • are moderately sensitive to changes in hydrology, patterns of inundation, discharge rates and/or human disturbance. • perform a moderate degree of water quality enhancement, but are insensitive to sustained eutrophication and/or pollution; and • are of importance for active and passive recreational activities.		>1 <= 2	C
Low/marginal Representative of wetlands that: • contain large areas of coarse (reeds) wetland vegetation with minimal floral and faunal diversity. • have a high urban watershed: wetland area ratio. • are important for active and passive recreation. • provide moderate to high levels of hydraulic buffering. • may be eutrophic and generally insensitive to further nutrient loading. • are generally insensitive to changes in hydrology, patterns of inundation, discharge rates and/or human disturbance. • have regulated water; and • contain large quantities of accumulated organic and inorganic sediments.		>0 <= 1	D
Rating	Explanation		
None, Rating = 0	Rarely sensitive to changes in water quality/hydrological regime		
Low, Rating =1	One or a few elements sensitive to changes in water quality/hydrological regime		
Moderate, Rating =2	Some elements sensitive to changes in water quality/hydrological regime		

Importance class (one or more attributes may apply)		Range of Median	Wetland Importance Class
High, Rating =3	Many elements sensitive to changes in water quality/ hydrological regime		
Very high, Rating =4	Very many elements sensitive to changes in water quality/ hydrological regime		

Freshwater classification

Riparian habitat includes:

“The physical structure and associated vegetation of areas associated with a watercourse which are commonly characterized by alluvial soil, and which are inundated or flooded to an extent and with a frequency sufficient to support vegetation of species with a composition and physical structure distinct from those of adjacent land areas”.

Aquatic ecosystem means: an ecosystem that is permanently or periodically inundated by flowing or standing water, or which has soils that are permanently or periodically saturated within 0.5 m of the soil surface.

Based on these definitions, for the purpose of the Classification System in this report, wetlands are a type of aquatic ecosystem because it is the presence of water at some stage (either permanently or periodically, sometimes rather ephemerally) that distinguishes a wetland ecosystem from a terrestrial ecosystem. Besides wetlands, as defined above, aquatic ecosystems are taken to also include rivers; lakes, ponds, dams and other open waterbodies; estuaries; and (shallow) marine systems. In terms of the legal definition (National Water Act, 1998), it is sometimes difficult to determine whether a particular aquatic ecosystem is a ‘wetland’. This does not hamper the use of the Classification System in this report, because no such distinction is made in the application of the Classification System.

In essence, the ecosystems included in the Classification System (i.e. all aquatic ecosystems, including wetlands, except for deep marine systems) encompass those that the Ramsar Convention defines, rather broadly, as ‘wetlands’, namely, “areas of marsh, fen, peatland or water, whether natural or artificial, permanent or temporary, with water that is static or flowing, fresh, brackish or salt, including areas of marine water the depth of which at low tide does not exceed six metres” (cited by Ramsar Convention Secretariat 2011).

South Africa is a geologically, geomorphologically, climatically, and ecologically complex country, and this has resulted in a diverse range of ecosystems, including rivers and wetland areas. To classify these aquatic ecosystems, this report used the Classification System for Wetlands and other Aquatic Ecosystems in South Africa User Manual (Olis *et al*; 2013).

This classification system is a hierarchical system that classifies all aquatic systems into levels as per Table 2.4:

Table 2.4: Aquatic system classification levels

Level 1: System setting	Inland systems		Estuarine systems		Marine systems	
Level 2: Regional setting	DWS Ecoregions		NFEPA Wetveg Groups		Other spatial frameworks	
Level 3: Landscape units	Valley floor	Slope		Plain		Bench

Level 4: Hydrogeomorphic (HGM) unit	River	Floodplain wetland	Channelled valley- bottom wetland	Unchannelled valley-bottom wetland	Depression	Seep	Wetland flat
Level 5: Hydrological regime	Perennially		Period and depth of inundation		Period of saturation		
Level 6: Descriptors	Natural vs Artificial	Salinity	pH	Substratum type	Vegetation cover type	Geology	

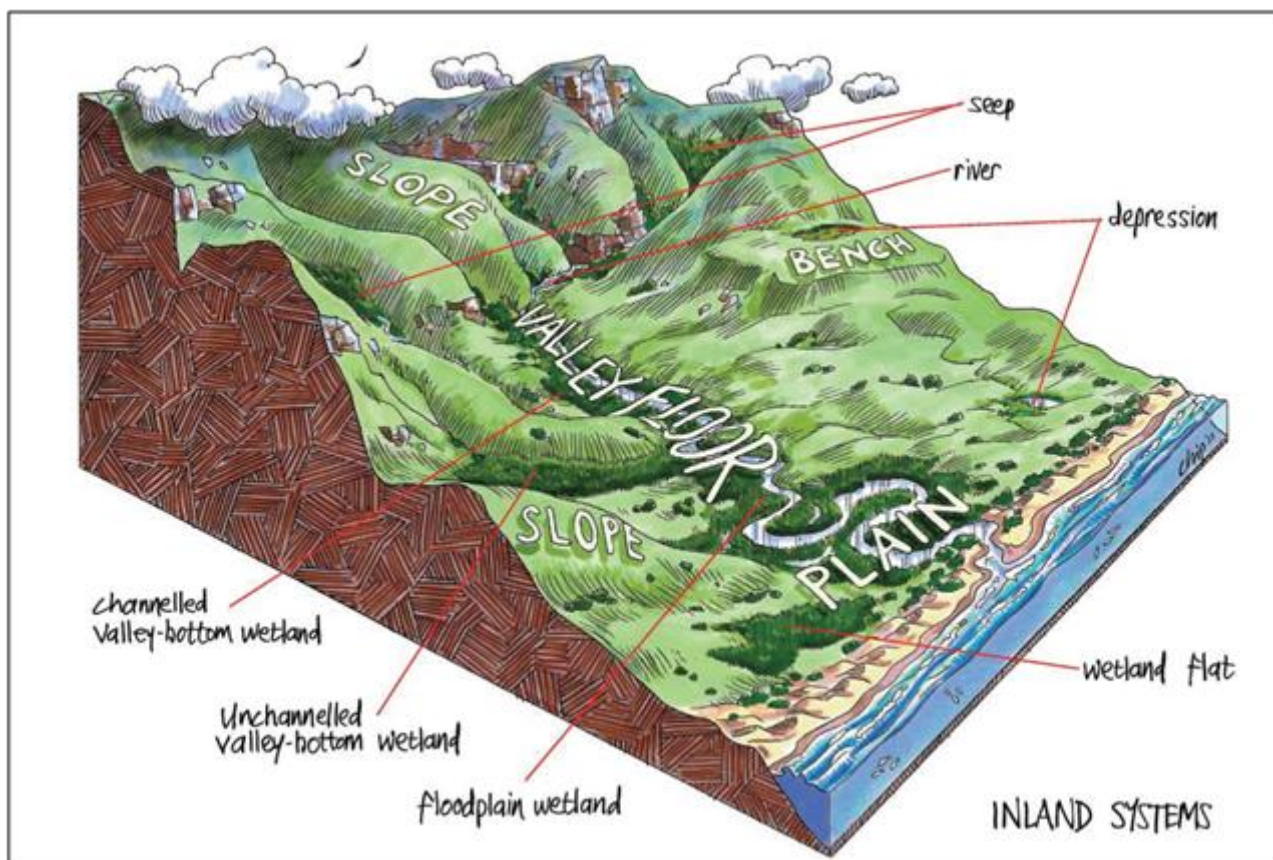


Figure 2.2: Illustration of the seven primary HGM units and their typical landscape settings (Ollis et.al, 2013).

2.3. Aquatic Biodiversity

Critical Biodiversity Areas (CBAs) are features critical for the conservation of biodiversity and maintenance of ecosystem functioning and should remain in a natural state as far as possible. CBAs also include freshwater components.

To assist in the development of these CBAs, each planning unit was classified by C-plan and Marxan based on a combination of factors including connectivity, habitat condition and presence of Red List Threatened species.

All features were grouped into the following aquatic CBA categories as listed in the Eastern Cape Biodiversity Conservation Plan (ECBCP, 2019):

Table 2.5: Critical biodiversity area categories as per the Eastern Cape Biodiversity Conservation Plan (2019)

Mapping Category	Critical Biodiversity Area Name	Land management objective
Protected areas (PA)	Protected Areas are areas that have been formally declared or recognized in terms of NEMPAA. This refers to “ <i>State owned</i> ” reserves, which includes National PAs managed by SANParks, Provincial PAs managed by Eastern Cape Parks and Tourism Agency, municipal reserves, Private Nature Reserves, and Protected Environments. Several municipal and provincial reserves have not been formally proclaimed under any legislation but are zoned accordingly in relevant Spatial Development Frameworks and are recognized as de facto Protected Areas.	Protected Areas are managed through Protected Area Management Plans and are therefore not managed through the ECBCP2019.
Critical Biodiversity Area (CBA)	CBA's are selected to meet biodiversity targets for species, ecosystems, and ecological processes. These include: Critically Endangered and Endangered Ecosystems. Critical linkage points (bottlenecks or pinch-points) in the corridor network; and All areas required to meet biodiversity targets and to ensure future persistence of species, ecosystems, and special habitats. CBA's are areas of high biodiversity value and should therefore be maintained in a natural state, with no further loss of habitat.	Maintain in a natural state (or near-natural state if this is the current condition of the site) that secures the retention of biodiversity pattern and ecological processes: For areas classified as CBA1, the following objectives must apply: ➤ Ecosystems and species must remain intact and undisturbed. ➤ Since these areas demonstrate high irreplaceability, if disturbed or lost, biodiversity targets will not be met. ➤ Important: these biodiversity features are at, or beyond, their limits of acceptable change. If land use activities are unavoidable in these areas and depending on expert opinion of the condition of the site, a Biodiversity Offset must be designed and implemented. For areas classified as CBA2, the following objectives must apply: ➤ Ecosystems and species must remain intact and undisturbed. ➤ There is some flexibility in the landscape to achieve biodiversity targets in these areas. It must be noted that the loss of a CBA2 area may elevate other CBA 2 areas to a CBA 1 category. ➤ These biodiversity features are at risk of reaching their limits of acceptable change. If land use activities are unavoidable in these areas, and depending on the condition of the site, set-aside areas must be designed in the layout and implemented. If site specific data confirms that biodiversity is significant, unique and/or highly threatened or that a Critically Endangered or Endangered species is present, Biodiversity Offsets must be implemented.
Ecological Support Area (ESA)	ESAs are areas NOT essential for meeting biodiversity targets, but are essential in terms of: Terrestrial landscape: Ensuring connectivity between CBA's, strengthening climate change resilience, and proper function of ecosystem infrastructure for delivery of ecosystem services. From a terrestrial perspective, ESAs may include riparian areas, coastal corridors, ridges, etc. Aquatic landscape: ESAs extend into catchments that are essential for the maintenance of CBA rivers and wetlands.	Maintain ecological function within the localized and broader landscape. A functional state in this context means that the area must be maintained in a semi-natural state such that ecological function and ecosystem services are maintained. For areas classified as ESA1, the following objectives apply: ➤ These areas are not required to

Mapping Category	Critical Biodiversity Area Name	Land management objective
	ESAs need to be maintained in a semi-natural, if not natural, state.	meet biodiversity targets, but they still perform essential roles in terms of connectivity, ecosystem service delivery and climate change resilience. ➤ These systems may vary in condition and maintaining function is the main objective, therefore: ○ Ecosystems still in natural, near natural state should be maintained. ○ Ecosystems that are moderately disturbed or degraded should be restored.
		Maintain current land use with no intensification. For areas classified as ESA2, the following objectives apply: ➤ These areas have already been subjected to severe and/or irreversible modification. ➤ These areas are not required to meet biodiversity targets, but they may still perform some function with respect to connectivity, ecosystem service delivery and climate change resilience. ➤ Objective is to maintain remaining function, therefore: ○ Areas should not undergo any further deterioration in ecological function. ○ Opportunities to change land use practices to improve ecological function (i.e. cultivation agriculture to livestock grazing agriculture) are desirable in ESA2 areas.
Other Natural Areas (ONA)	ONAs are areas in a natural or near natural state that have not been identified as priority areas in the current plan. ONAs still support biodiversity and deliver ecosystem services.	No desired state or management objective is provided for ONA or NNR.
No Natural Areas Remaining (NNR)	NNR areas that are heavily, permanently, modified and are no longer considered natural. Although some biodiversity and ecological function may be retained, irreversible impacts on biodiversity mean that they cannot contribute towards targets	

2.4. Environmental Screening Tool

The National Web based Environmental Screening Tool is a geographically based web-enabled application which allows a proponent intending to apply for environmental authorization in terms of the Environmental Impact Assessment (EIA) Regulations 2014, as amended to screen their proposed site for any environmental sensitivity.

The Screening Tool also provides site specific EIA process and review information, for example, the Screening Tool may identify if an industrial development zone, minimum information requirement, Environmental Management Framework or bio-regional plan applies to a specific area.

Some of these documents can then be accessed through the Screening Tool via links, for consideration during screening.

Further to this, the Screening Tool identifies related exclusions and/ or specific requirements including specialist studies applicable to the proposed site and/or development, based on the national sector classification and the environmental sensitivity of the site.

Finally, the Screening Tool allows for the generating of a Screening Report referred to in Regulation 16(1)(v) of the Environmental Impact Assessment Regulations 2014, as amended whereby a Screening Report is required to accompany any application for Environmental Authorization and as such the tool has been developed in a manner that is user friendly and no specific software or specialized GIS skills are required to operate this system.

2.5. Site Sensitivity

The screening tool has identified species and ecosystem spatial triggers likely to indicate environmental sensitivity associated with a particular proposed development site, which in turn determined the necessity and requirements for conducting a Terrestrial Biodiversity Assessment.

Section 5 of this report identified and mapped zones of Site Ecological Sensitivity (SEI) within each PAOI. These zones are based on initial sensitivity identification in the Screening Report, followed by a site inspection of the entire PAOI and a detailed assessment of the area.

2.6. Risk Assessment Protocol

The risk assessment process follows the DWS steps as listed in Appendix A of GNR 509 of 2016 (General Authorisation in terms of section 39 of the National Water Act, 1998 (Act No. 36 of 1998) for water uses as defined in section 21(c) or section 21(i).

Risk is determined after considering all listed control/mitigation measures. Borderline Low/Moderate risk scores can be manually adapted downwards up to a maximum of 25 points (from a score of 80) subject to listing of additional mitigation measures considered and listed.

Only low risk activities located within the regulated area of the watercourse will qualify for a GA. Medium and High-risk activities will require a Section 21 (c) & (i) water use license.

The following criteria based on the DWS 2023 publication: Section 21 c and i water use Risk Assessment Protocol:

Table 2.6. Ecological importance of affected watercourse/s

What is the ecological importance of the watercourse/s, based on the criteria and guidelines provided below?

Low EIS; OESA's; presence of species/habitats of Least Concern on the IUCN Red List or on a regional/national Red List and/or species which are common and widespread and/or habitats of low conservation interest	Low = 2
Medium EIS; ESAs; presence of species/habitats listed as VU or NT on the IUCN Red List; or listed as VU or NT on a regional/national Red List	Moderate = 3
High EIS; CBA2; presence of species/habitats listed as EN or CR on the IUCN Red List; or listed as EN or CR on a regional/national Red List	High = 4
Very high EIS; CBA1, NFEPA; presence of species/habitats listed as EN or CR on the IUCN Red List; or listed as EN or CR on a regional /national Red List; KBA or IBA's	Very high = 5

EIS=Ecological Importance and Sensitivity; OESA=Other Ecological Support Areas; IUCN=International Union

for Conservation of Nature; ESA=Ecological Support Areas; VU=Vulnerable; NT=Near Threatened; EN=Endangered; CR=Critically Endangered; CBA=Critical Biodiversity Area; NFEPA=National Freshwater Priority Area; KBA=Key Biodiversity Area; IBA=Important Bird Area.

Table 2.7. Intensity of Impact

What is the intensity of the impact on the resource quality (hydrology, water quality, geomorphology, biota)?

Negative Impacts	
Negligible / non-harmful; no change in PES	0
Very low / potentially harmful; negligible deterioration in PES (<5% change)	+1
Low / slightly harmful; minor deterioration in PES (<10% change)	+2
Medium / moderately harmful; moderate deterioration in PES (>10% change)	+3
High / severely harmful; large deterioration in PES (by one class or more)	+4
Very high / critically harmful; critical deterioration in PES (to E/F or F class)	+5
Positive Impacts	
Negligible; no change in PES	0
Very low / potentially beneficial; negligible improvement in PES (<5% change)	-1
Low / slightly beneficial; minor improvement in PES (<10% change)	-2
Medium / moderately beneficial; moderate improvement in PES (>10% change)	-3
Highly beneficial; large improvement in PES (by one class or more) and/or increase in protection status	-4
Very highly beneficial; improvement to near-natural state (A or A/B class) and/or major increase in protection status	-5

NOTE: Positive Impacts must be given a negative intensity score

***PES of affected watercourses must be considered when scoring Impact Intensity**

Table 2.8: Spatial scale (extent) of impact

How big is the area that the activity is impacting on, relative to the size of the impacted watercourses?

Very small portion of the of watercourse/s impacted (<10% of extent)	1
Moderate portion of watercourse/s impacted (10-60% of extent)	2
Large portion of watercourse/s impacted (60-80%)	3
Most or all of watercourse/s impacted (>80%)	4
Impacts extend into watercourses located well beyond the footprint of the activities	5

Table 2.9: Duration

How long does the activity impact on the resource quality?

Transient (One day to one month)	1
Short-term (a few months to 5 years) OR repeated infrequently (e.g. annually) for one day to one month	2
Medium term (5-15 years)	3
Long-term (ceases with operational life)	4
Permanent	5

Table 2.10: Likelihood of the impact

What is the probability that the activity will impact on the resource quality?

Improbable/unlikely	20%
Low probability	40%
Medium probability	60%
High probable	80%
Definite/Unknown	100%

Table 2.11: Risk Rating classes

RATING	CLASS	MANAGEMENT DESCRIPTION
1 – 29	(L) Low Risk OR (+) Positive (++) Highly Positive	Acceptable as is or with proposed mitigation measures. Impact to watercourses and resource quality small and easily mitigated, or positive.

RATING	CLASS	MANAGEMENT DESCRIPTION
30 - 60	(M) Moderate Risk	Risk and impact on watercourses are notable and require mitigation measures on a higher level, which costs more and require specialist input. Licence required.
61 – 100	(H) High Risk	Watercourse(s) impacts by the activity are such that they impose a long-term threat on a large scale and lowering of the Reserve. Licence required.

Table 2.12: Calculations and maximum values

Intensity = Maximum Intensity Score (negative value for positive impact)	MAX = 5
Severity = Intensity + Spatial Scale + Duration (<Intensity - Spatial Scale - Duration> for positive impact)	MAX = 15 (MIN = -15 for +ve impacts)
Consequence = Severity X Importance rating	MAX = 75
Significance\Risk = (Consequence X Likelihood) X (100/75)	MAX = 100

The risk matrix register is also based on the DWS 2023 publication: Section 21 c&i water use Risk Assessment Protocol. Refer to Appendix A of this report of the detailed Risk Matrix register.

2.7. Buffer determination

The DWS as part of the Water Resource Classification System (WRCS) mandates a specific procedure for identifying and managing water features within a given area. This involves several steps and considerations:

- Delineation of watercourse features which includes all visible bodies of water such as wetlands, rivers, and streams, including all riparian habitats. In addition, flood lines represent the potential extent of flooding based on historical data or predictive models. They indicate areas that might be submerged during a flood event. If a flood-line was done, then the flood-line should be compared with the riparian habitat, and waterbody; the greatest of the three will be the defined watercourse.
- A buffer zone of 100 m must be established around rivers. This buffer serves as a protective barrier to maintain the ecological integrity of the watercourse and to mitigate the impact of adjacent activities. Whilst, wetlands, which are areas where water saturation is the primary factor influencing the soil and plant life, require a more extensive buffer of 500 m. This larger buffer is due to the sensitive and critical nature of wetlands in supporting biodiversity, water purification, and flood control.
- The delineated features and their respective buffer zones are designated as "Regulated Areas." These areas are subject to specific regulations and monitoring to ensure their protection. Activities within these Regulated Areas are scrutinised to determine if they might pose a risk to the water features.
- Risk Assessment Matrix (RAM): The Risk Matrix is a tool used to assess the potential impact of proposed activities within the Regulated Areas. Depending on the level of risk, the matrix will indicate whether a General Authorisation (GA) or a full Water Use License is required.
 - General Authorisation (GA): Typically required for activities with a lower risk and smaller impact on water resources.
 - Full License: Required for higher-risk activities that have a significant potential to affect water quality or quantity.

The DWS no longer uses other buffer delineation methods, such as one defined by Macfarlane et al. (2014). The focus is solely on the specified 100 m and 500 m buffers around rivers and wetlands, respectively.

2.8. Impact assessment

The impacts that may result from the planning and design phase, construction phase, operation phase of the proposed Walmer Housing development were assessed according to several criteria to arrive at an overall significance rating. The criteria used were as follows (based on DEAT 2002 - Impact Significance, IEM Information Series 5; and DEAT 2006 - Assessment of Alternatives and Impacts in support of the EIA Regulations, IEM Guideline Series 5):

Table 2.13: Criteria used in determining significance ratings to potential impacts

ASPECT	IMPACT RATING	
Status of the Impact	A statement of whether the impact is positive (a benefit), negative (a cost), or neutral	
Direct impact	Impacts that are caused directly by the activity and generally occur at the same time and at the place of the activity. These impacts are usually associated with the construction, operation or maintenance of an activity and are generally obvious and quantifiable.	
Indirect Impacts	Indirect impacts are not a direct result of the project but are often produced away from or because of a complex impact pathway related to the project.	
Cumulative Impacts	Impacts that result from the incremental impact of the proposed activity on a common resource when added to the impacts of the past, present, or reasonably foreseeable future activities. Cumulative impacts can occur from the collective impacts of individual minor actions over a period of time and can include both direct and indirect impacts.	
Nature of the Impact	The evaluation of nature is impact specific. Most negative impacts will remain negative, however, after mitigation, significance should be reduced to: ➤ Positive ➤ Negative	
Extent	A description of whether the impact would occur on a scale limited to within the study area (local), limited to within 5 km of the study area (area) on a regional scale. i.e. the Nelson Mandela Metro & Eastern Cape (Region); or would occur on a national or international scale.	
Duration	A prediction of whether the duration of the impact would be immediate and once-off (less than one month), more than once, but short term (less than one year), regular, medium term (1 to 5 years), long term (6 to 15 years), project life/permanent (> 15 years, with the impact ceasing after the operational life of the development or should be considered as permanent).	

ASPECT	IMPACT RATING		
Intensity	This provides an order of magnitude of whether or not the intensity (magnitude/size/frequency) of the impact would be negligible, low, medium, high or very high. This is based on the following aspects: ➤ An assessment of the reversibility of the impact (permanent loss of resources, or impact is reversible after project life. ➤ Whether or not the aspect is controversial. ➤ An assessment of the irreplaceability of the resource loss caused by the activity (whether the project will destroy the resources which are easily replaceable, or the project will destroy the resources which are irreplaceable and cannot be replaced. ➤ The level of alteration to the natural system, processes or systems.		
	Negligible	The impact does not affect physical, biophysical or socio-economic functions and processes.	1
	Low/potential harmful	The impact has limited impacts on physical, biophysical or socio-economic functions and processes.	2
	Medium/slightly harmful	The impact has an effect on physical, biophysical or socio-economic functions and processes, but in such a way that these processes can still continue to function albeit in a modified fashion.	3
	High/harmful	Where the physical, biophysical or socio-economic functions and processes are impacted on in such a way as to cause them to temporarily or permanently cease.	4
	Very high/disastrous	Where the physical, biophysical or socio-economic functions and processes are highly impacted on in such a way as to cause them to permanently cease.	5

Based on a synthesis or combination of the information contained in the above-described criteria; and drawing on legal policies and guidelines as well as the status of the impacts and potential risks, the overall significance was determined as follows:

Table 2.14: Definition of significance ratings (positive and negative)

Significance	Description
Very high (VH) (150+)	An impact of very high significance will mean that the project cannot proceed, and the impacts are irreversible, regardless of available mitigation options.
High (H) (101-149)	An impact of high significance could influence a decision about whether to proceed with the proposed project, regardless of available mitigation options.
Medium (M) (51-100)	If left unmanaged, the impact of medium significance could influence a decision about whether to proceed with a proposed project.
Low (L) (25-50)	An impact of low significance would have little effect on decision making and only a small influence on project design or alternative motivation.
Very low (VL) (1-24)	An impact of very low significance is likely to contribute to positive decisions about whether to proceed with the project. It will have little effect and is unlikely to have an influence on project design or alternative motivation.
Negligible / zero impact	There will be no impact, or any impact identified can be viewed as negligible. This rating will be unlikely to have an influence on project design or alternative motivation.
Positive impact (+)	A positive impact is likely to result in a positive consequence/effect and is likely to contribute to positive decisions about whether to proceed with the project.

3. Site assessment

This chapter compares baseline information with field survey data collected. A site visit was conducted on the 10th June 2026. Data collected during the site visit was then compared to existing literature for the site which included vegetation classifications and biodiversity programs and plans.

3.1 Site Sensitivity Verification

According to the protocols specified in GN 1540 (Procedures for the Assessment and Minimum Criteria for Reporting on Identified Environmental Themes in Terms of Sections 24(5)(A) and (H) and 44 of the National Environmental Management Act, 1998, when Applying for Environmental Authorization), assessment and reporting requirements for aquatic biodiversity are associated with a level of environmental sensitivity identified by the national web-based environmental screening tool (screening tool). The Aquatic Biodiversity Theme Protocol was assessed in this report.

The proposed development area environmental sensitivity for the aquatic biodiversity theme was identified as **very high sensitivity** in the DFFE Screening Tool Report (Figure 3.1).



Figure 3.1: Map of the The DFFE Screening Tool map of the proposed housing development in Port Elizabeth, Eastern Cape.

Based on the site verification (conducted further below) the site sensitivity **will remain as very high sensitivity**. This is due to the following reasons:

1. Very High sensitivity for ESA1 areas occur in the Total PAOI.
2. Very High sensitivity for SWSA (sw)_Tsitsikamma.

The purpose of the initial site investigation was to confirm this sensitivity allocation. As per the Protocol for the Specialist Assessment and Minimum Report Content Requirements for Environmental Impacts on Aquatic Biodiversity the aquatic biodiversity theme must be assessed as a full assessment.

3.1. Topography

The project development site sits on a relatively flat to gently sloping hill or ridge. Elevation inside the main project boundary ranges primarily between 105 and 115 meters. The land generally slopes downwards toward the north and north-east.

Key Topographical Features

Highest Terrain: The highest points are located to the west and southwest of the site, where contours rise up to 125, 130, and 140 meters.

Lowest Terrain: The lowest points are found toward the north and northeast boundaries of the buffer zone, dropping down to 95 and 90 meters.

Drainage/Depressions: A subtle valley or lower drainage area (around the 100m to 105m contours) curves around the northern and eastern sides of the main development site.

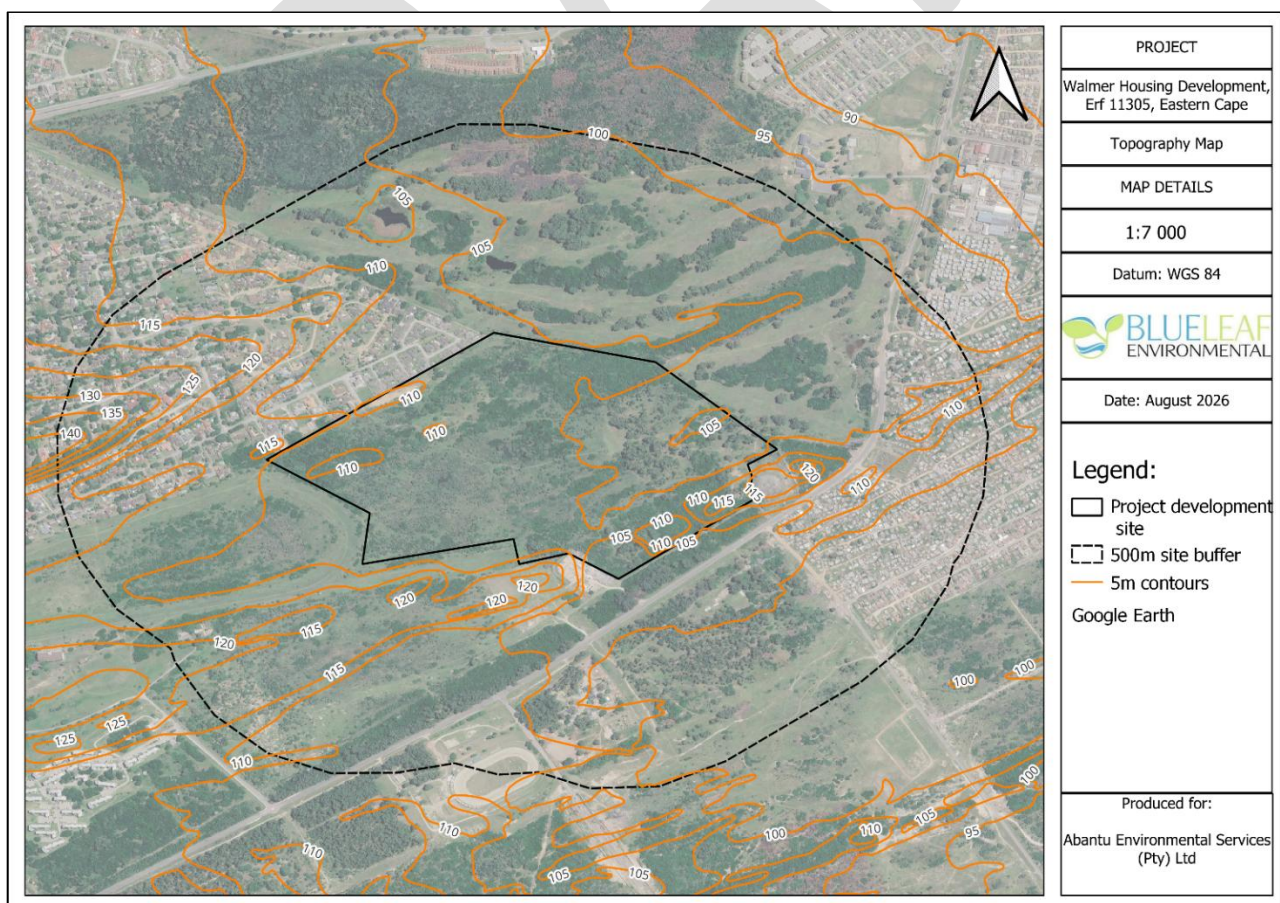


Figure 3.2: Contour map of the proposed project site and surrounding areas

3.2. Local climate

The climate is warm and temperate in Port Elizabeth. In Port Elizabeth, the average annual temperature is 18.1 °C. The rainfall here is around 563 mm per year. Port Elizabeth is situated in the southern hemisphere. Summer begins here at the end of January and ends in December. The most popular time to visit is January, February. The driest month is January, with 39 mm of rainfall. The greatest amount of precipitation occurs in November, with an average of 58 mm. The warmest month of the year is February, with an average temperature of 22.1 °C. The lowest average temperatures in the year occur in July, when it is around 14.4 °C.

3.3. Geology and Soils

The geology of the project area is classified as the Schelm Hoek Formation. These formations are located within the larger Algoa Group. The Algoa Group includes marine terrace conglomerate, coquinite, aeolianite, alluvial gravel, sand, silt, calc-tufa and minor gravel.

The Schelm Hoek Formation is the most recent of all the geological formations, overlying the Nanaga, Salnova and Nahoon Formation rocks. It consists of unconsolidated calcareous sand (coastal dunes) and minor palaeosols (Cape Farm Mapper, v. 3). The unconsolidated calcareous sand consists of wind-blown (aeolian) sand on the present-day coastline, which is still being deposited.

3.4. Land use

A site inspection was conducted on the 27th of July 2026.

The greatest portion of the site or Primary PAOI consists of green open space i.e. grassland that is invaded by Australian Rooikrans and Port Jackson Willow trees – with a few remnant patches of partially intact forest/thicket vegetation, surrounded by Sweet Thorn (*Vachellia karroo*) trees, present in the northwestern, northern and southeastern portions. The site currently has no specific land use other than green open space or vacant and unused land. The site is used to for the illegal dumping of household waste, garden waste and building rubble; to burn waste and cut firewood, and as a thoroughfare by local residences to travel between suburbs.

According to Google Earth aerial imagery, there is a ruin of an old building or house in the southeastern portion of the site, just west of the reservoir. This building/house was still standing or in use in 2004, after which it was partially demolished. There is also a large soil/overburden mound in the southern portion of the site, northeast of SmartStone Port Elizabeth, with Eucalyptus trees growing on and around it. The reason for its presence is unclear.

An old, dilapidated observation tower of the old Arlington Racecourse is present on the site boundary in the westernmost portion of the site. An area in the central southwestern portion of the site, nearest to the Beaumont Estate, was cultivated until 2018 (at minimum), judging by the presence cultivated blocks separated by hedges and straight furrows in the ground. An area south of the residences in Beaumont Estate, along the central northwestern site boundary, was either cleared of alien invasive trees and levelled, or cleared, excavated and mined for building sand, in the latter half of 2017 (as evidenced by the presence of an area of bare and levelled ground in December 2017). Similarly, a small area of ground in the southeastern portion of the site was cleared of grasses and/or excavated of topsoil at the end of 2010.

The green open space on site is classed as contiguous low forest & thicket, dense forest & woodland, smallholdings (tree), smallholdings (low veg/grass), smallholdings (bush), urban recreational fields (tree) and urban recreational fields (grass) (see Figure 3.3, below; Cape Farm Mapper, v. 3). Developed areas i.e. residences and businesses/industries, are classed residential formal (tree), residential formal (low veg/grass), village scattered (bare & low veg/grass combo) and industrial. Very limited areas are classed low shrubland (fynbos) and natural grassland.

Land uses of the Secondary PAOI include: built-up areas i.e. dirt and tarred roads, low-cost and informal housing, free-standing residences, businesses, estates and townhouse complexes; green open space i.e. grassland with predominantly alien and invasive Acacia trees, a continuous band of forest/thornveld vegetation, an abandoned golf course (with artificial wetlands) and horse racecourse with grassland and patches of forest/thornveld; space), and service infrastructure (reservoir, powerlines, telephone lines).

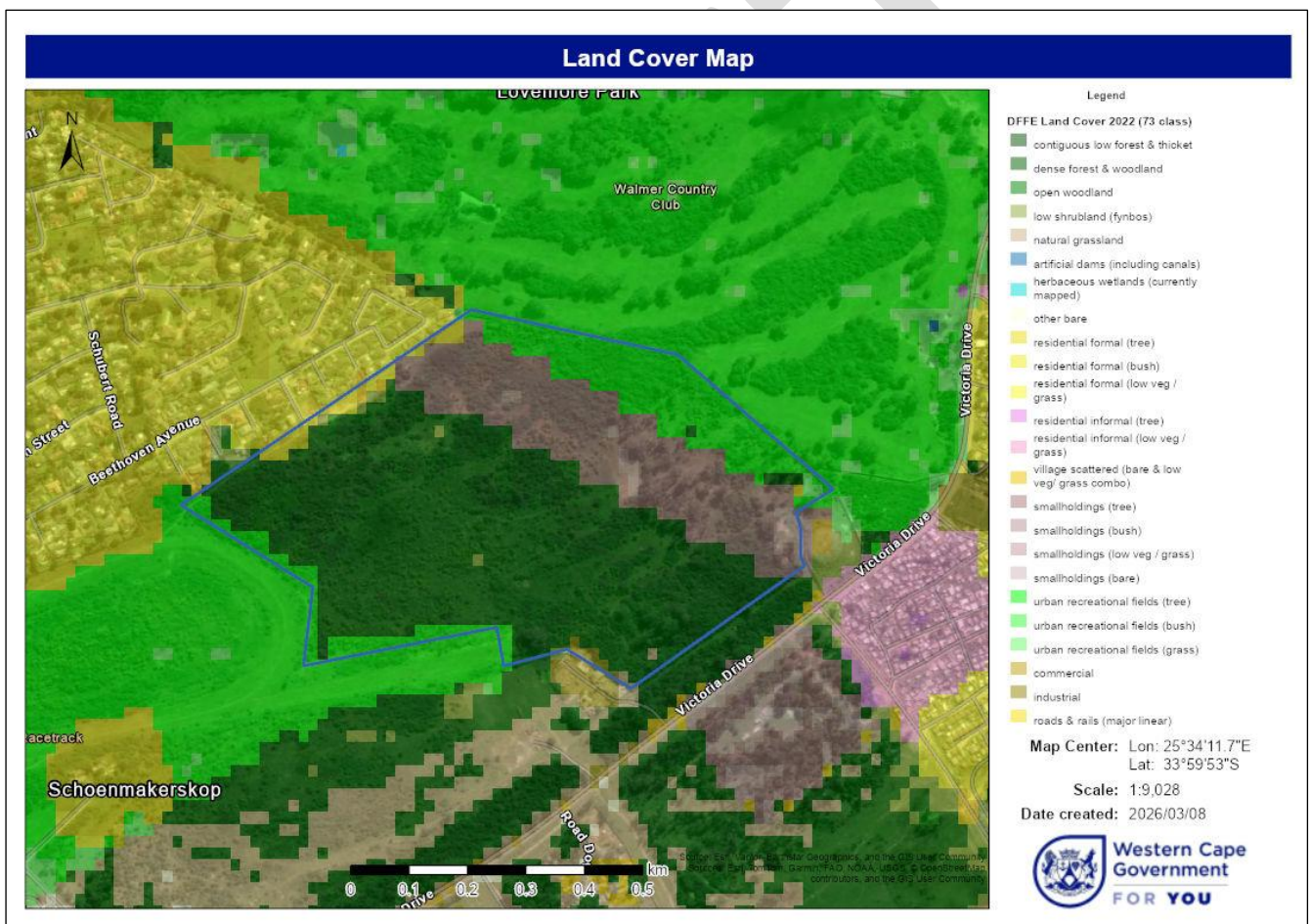


Figure 3.3: Land cover map of the proposed project sites and surrounding areas.

Below are photographs illustrating the land cover and land uses for the study site and surrounding areas.

Reservoir surrounded by a concrete wall, located east of the site and north of Victoria Drive:



Dense forest/thicket vegetation located on Victoria Drive – southeast of the site and southwest of the reservoir – with a shack and an area that is used to sort and burn waste (presumably from the nearby Arlington Waste Facility):



Dense forest/thicket vegetation on Victoria Drive – southeast of the site:



Area of raised ground/soil or overburden heap with Eucalyptus and Port Jackson Willow trees in the southern/southeastern portion of the site:



Excavated area south of the residences at Beaumont Estate – on the central northern site boundary (facing east):



Facing west on the dirt road that runs along the northern site boundary – residences of Beaumont Estate to right in photograph:



Walking path bordered by grassland, thicket-type shrubs and alien invasive trees in the central northwestern portion of the site:



Dumped garden and household waste surrounded by invaded grassland:



Patches of invasive Castor-oil Plant (*Ricinus communis*) amid grassland, with invasive Port Jackson Willow and indigenous Sweet Thorn trees present along the edge of forest/thicket patches:





Forest/thicket patch in the central portion of site:



Forest/thicket vegetation in the southern portion the site:





Chopped branches of Rooikrans trees amid invaded grassland:



Observation tower on the western site boundary at the old Arlington Racecourse – racecourse to right in photograph:



Forest/thicket patches, Port Jackson Willow and Eucalyptus trees in the southern portion of the site:



3.5. Vegetation

Site vegetation was first described according to available literature and databases. A site visit was then conducted to confirm and describe vegetation distribution as well as the condition of each vegetation unit identified.

According to SANBI (Vegetation Map, updated in 2024), the Primary PAOI is located within two vegetation types, namely: **Algoa Sandstone Fynbos (FFs 29)** and **Sardinia Forest Thicket (AT 48)**.

Algoa Sandstone Fynbos:

Distribution: Eastern Cape Province: Coastal flats at Port Elizabeth from Van Stadens River in the west to Southdene- Summerstrand in the east, located mostly some kilometres from the coast and close to the coast at only Maitland River Mouth and urbanized Summerstrand. Altitude 20–300 m.

Vegetation & Landscape Features: Flat to slightly undulating plain supporting grassy shrubland (mainly graminoid fynbos). Grass has become dominant, especially in wet habitats. In the south this fynbos unit borders on AT 9 Albany Coastal Belt and AZs 1 Algoa Dune Strandveld and forms transitional mosaics with both. It also borders on patches of FOz 6 Southern Coastal Forest in this area.

SANBI classifies Algoa Sandstone Fynbos (FFs 29) as **Endangered** with some of the targeted 23% statutorily conserved in the Van Stadens Wildflower Reserve, The Island Nature Reserve as well as in several private nature reserves. More than 50% transformed (cultivation, urban sprawl of the Nelson Mandela Metropolitan Area). Several Australian Acacia species occur as invasive aliens, but only to a limited extent. Erosion moderate and very low.

Sardinia Forest Thicket (AT 48)

Distribution: This vegetation type occurs in the Eastern Cape Province. South-east of Port Elizabeth, forming an arc from Seaview in the west to Walmer Heights in the east, and occurring within 2-5 km of the coast.

Vegetation & Landscape Features: Wind- and fire-protected areas on moderately undulating coastal

dunes. Medium-sized to tall (3 - 5 m) dense thicket, dominated by small trees and woody shrubs, intergrading with small pockets of tree-dominated (*Ekebergia capensis*, *Erythrina caffra*) coastal forest.

SANBI classifies Sardinia Forest Thicket (AT 48) as **Critically Endangered** with some of the targeted 19% statutorily conserved in areas not listed. 35.99% transformed due to Cultivation, mining, overgrazing. Erosion low.

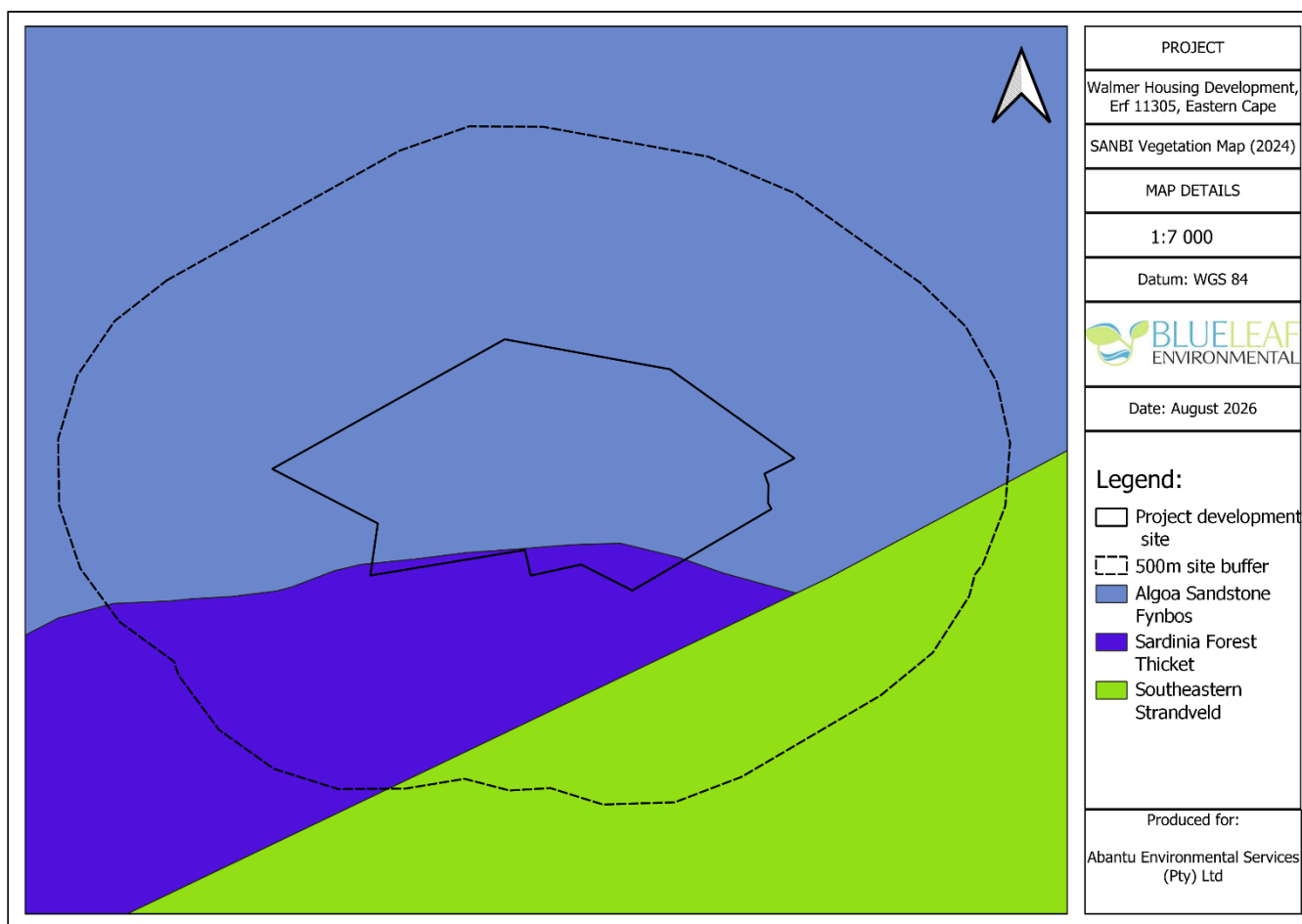


Figure 3.4: SANBI Vegetation map of the proposed project sites and surrounding areas.

3.6 Aquatic biodiversity

Freshwater ecosystems, despite covering less than 1% of the Earth's surface, are biodiversity hotspots, supporting at least 10% of Earth's species, including a wide array of plants, animals, and microorganisms. Freshwater habitats, including rivers, lakes, and wetlands, are incredibly diverse, with more than 400 large-scale ecoregions and thousands of individual water bodies.

Freshwater ecosystems provide vital ecosystem services, including clean water, food, and habitat for various species. They are crucial for human well-being, providing water for drinking, agriculture, and recreation, as well as food and economic opportunities.

Freshwater biodiversity is under increasing threat from various factors, including pollution, habitat degradation, climate change, and overexploitation.

3.6.1. Freshwater identification

Based on the above, no development is recommended in any freshwater area without approval for the competent authority (DWS). Various databases of freshwater systems exist in SA. Below is a summary of each database findings with a summary of what was found during the site verification.

National Wetland Map 5:

The National Wetland Map version 5 (NWM5) shows the distribution of inland wetland ecosystem types across South Africa and includes estuaries and the extent of some rivers. Further to this, a confidence map was compiled to identify areas where wetland extent and hydrogeomorphic (HGM) units (which contributed to defining the inland wetland ecosystem types together with the regional setting) attained at a higher level of certainty compared to other areas.

Five levels are listed. Higher levels of certainty are associated with areas that have been visited in-field by a wetland specialist(s) over multiple seasons and cycles of the wetland hydroperiod and are therefore more accurately represented in the dataset.

Table 3.1: Confidence ratings assigned to sub-quaternary catchments based for inland wetlands (NWA 2018)

Rating	Description
1 – Low	Desktop mapping of the extent of inland wetlands was done by non-wetland specialists for a part of or to the full extent of the sub-quaternary catchment (SQ4) (Mostly DRDLR: NGI for non-wetland typing purposes).
2 – Low to medium	Desktop mapping of inland wetland extent was captured by interns for the purpose of NWM5 to the full extent or a part of the SQ4. This may also include areas where data from wetland specialists had been incorporated, but the dataset was either not typed to the HGM unit, or complete for all HGM units, or was based on old imagery.
3 – Medium	Desktop mapping of the extent of inland wetlands and HGM typing was done by wetland specialists for the full extent of the SQ4.
4 – Medium to high	Desktop mapping of the extent of inland wetlands and HGM typing, as well as field verification and revision by experts was completed for the full extent of the SQ4.
5 – High	Inland wetlands have been mapped and verified for a period of > 10 years over multiple hydrological cycles/ hydroperiod for the full extent of the SQ4. Verification may include field observations as well as soil and/or vegetation surveys.

According to the NWM5 database, the site has been classified as having a Wetland Confidence of 3. Since specialists have been to site for field verification and revision, the Wetland Confidence has thus changed to 4, indicating a **Medium – High level of confidence** on the accuracy that the extent and HGM unit are well represented.

According to the NWM5 database, 11 wetlands can be seen within the Total PAOI (Figure 3.5). Refer to the Surface Water map (Figure 3.8) below for a final, high confidence surface water map of the Total PAOI.

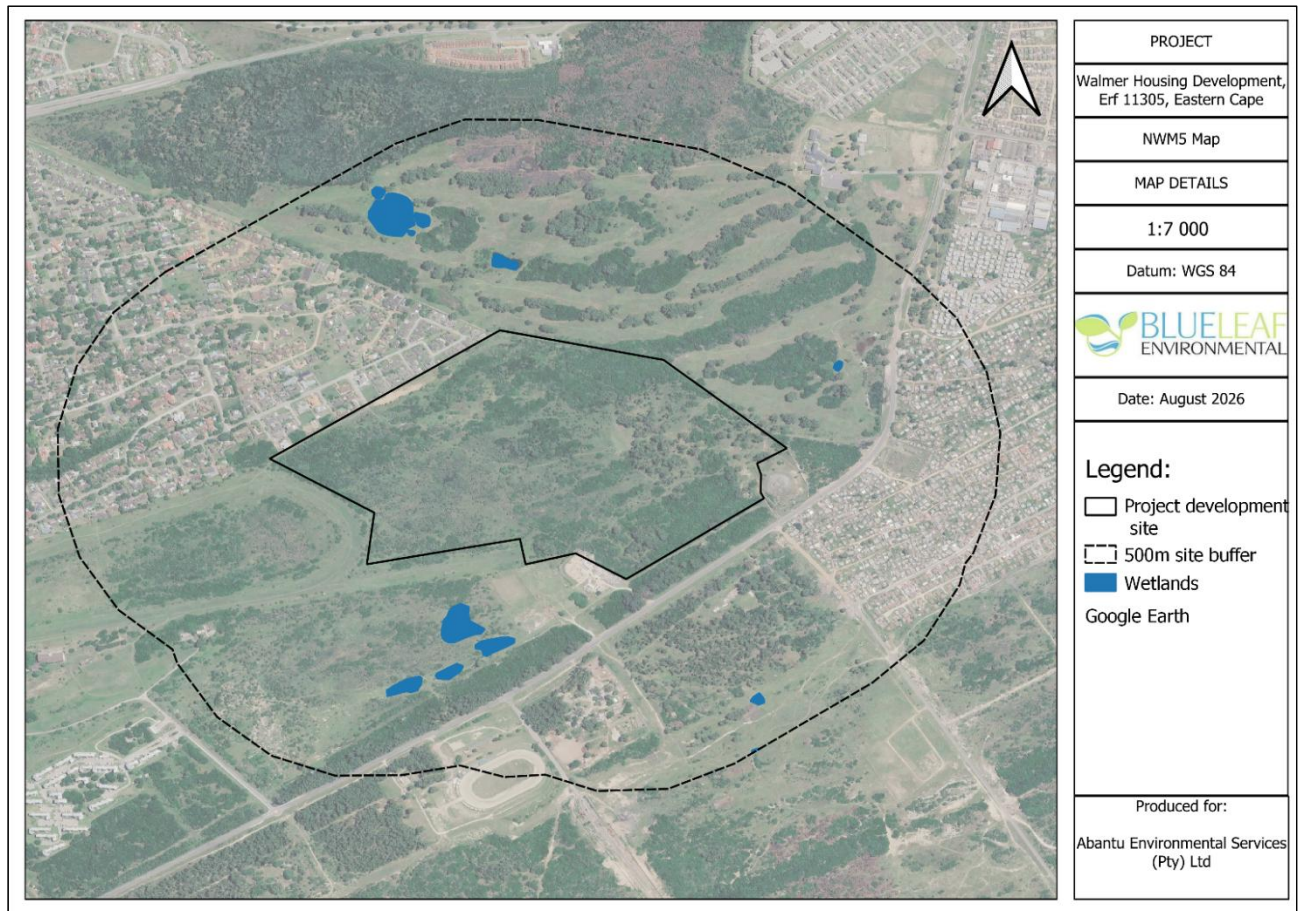


Figure 3.5: NWM5 map of the proposed project site and surrounding areas

NFEPA Wetlands:

Wetland National Freshwater Priority Areas (FEPAs) are wetland ecosystem types and wetland conditions on a national scale. The delineations were based largely on remotely sensed imagery and therefore did not include historic wetlands lost through drainage, ploughing and concreting. Irreversible loss of wetlands is expected to be high in some areas, such as urban centres. In addition, there are many gaps in wetlands as remote sensing does not detect all wetlands.

According to the NFEPA database, only one wetland was observed within the Total PAOI. A site inspection has confirmed that this wetland is actually a reservoir and should be classified as such. NFEPA wetlands are classified differently from the NWM5 map but will be properly classified in this assessment further within this report. Again, refer to the surface water map (Figure 3.8) below for a final, high confidence surface water map of the Total PAOI.

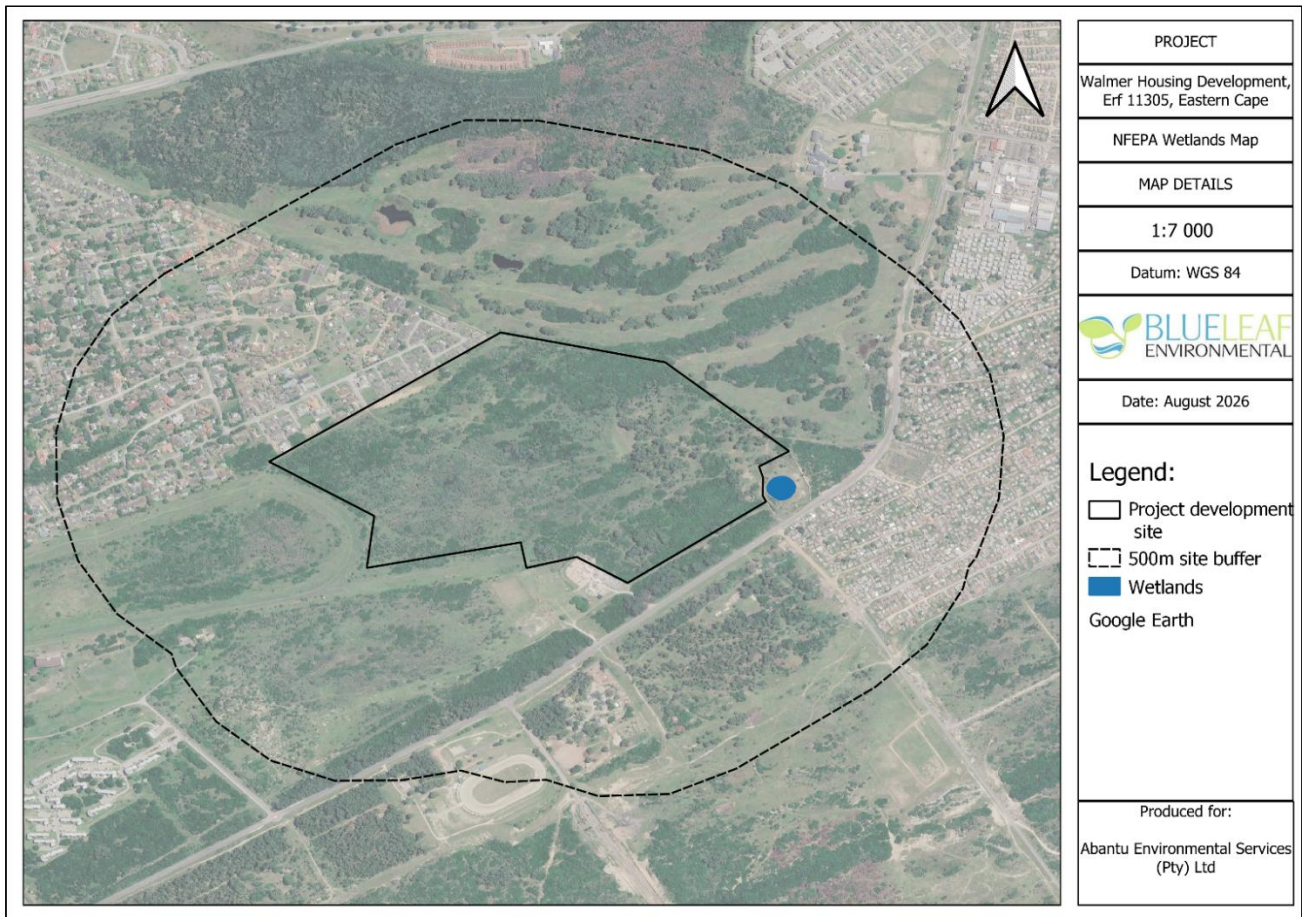


Figure 3.6: NFEPA wetlands map of the proposed project site and surrounding areas

NFEPA Rivers:

According to the NFEPA River database, no rivers were identified within the Total PAOI. Refer to the Surface Water maps (Figure 3.8) below for final, high confidence surface water maps of the Total PAOI. Rivers identified are delineated, classified and discussed in section 4 below.

NBA 2018 Rivers

According to the NBA (National Biodiversity Assessment, 2018) database for rivers, no rivers were identified within the Total PAOI of the project development site. The Baakens River can be observed north of the project development site which falls outside of the Total PAOI. Refer to the Surface Water map (3.8) below for final, high confidence maps of the Total PAOI.

3.6.2. Eastern Cape Biodiversity Conservation Plan (ECBCP)

This section only addresses aquatic related biodiversity sensitive areas. Terrestrial biodiversity sensitive area is addressed in the Terrestrial Biodiversity Assessment for the project.

According to the Eastern Cape Biodiversity Conservation Plan – Freshwater Biodiversity (ECBCP2019), north of the proposed development site occurs within an Aquatic Ecological Support Area (ESA1) (Fig 3.7).

The sensitivity allocation is due to the ESA1 functioning as vital ecological infrastructure under the Eastern Cape Biodiversity Conservation Plan. The zone directly north of Erf 11305 contains the

Baakens River valley system and its local drainage paths. In a freshwater conservation framework, these areas are flagged to shield water source areas, manage stormwater runoff, and control soil erosion from urban encroachment.

ESA's are important in supporting the functioning of CBAs and are often vital for delivering ecosystem services. ESA 1 are areas in a natural condition that are required to meet biodiversity targets, for species, ecosystems or ecological processes and infrastructure. ESA 2 are areas that are not essential for meeting biodiversity targets but are targeted for restoration and play an important role in supporting the functioning of Protected Areas (PAs) or CBAs and are often vital for delivering ecosystem services.

The site does not fall within a CBA2.

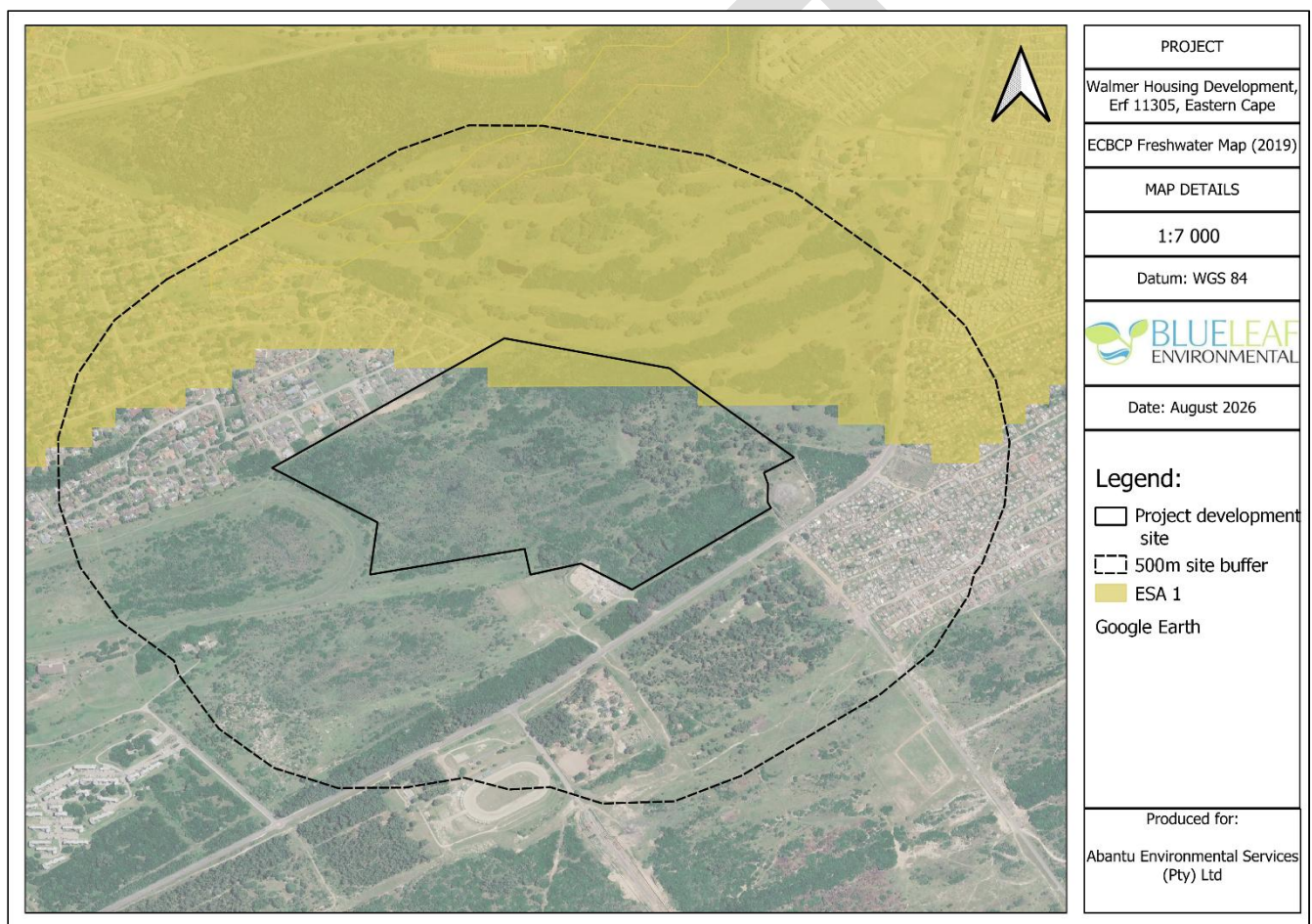


Figure 3.7: ECBCP Freshwater map of the project site and surroundings

3.6.3. Surface water

The following freshwater features were identified within the Total PAOI showing the following water feature characteristics (Fig 3.8):

- All water courses (including rivers, springs, natural channels in which water flows regularly or intermittently).
- All wetlands (including lakes or dams into which, or from which, water flows).
- Any collection of water which the Minister may, by notice in the Gazette, declare to be a watercourse, and a reference to a watercourse include, where relevant, its bed and banks.

- All wetlands and rivers identified during the site visit but not appearing within any database.

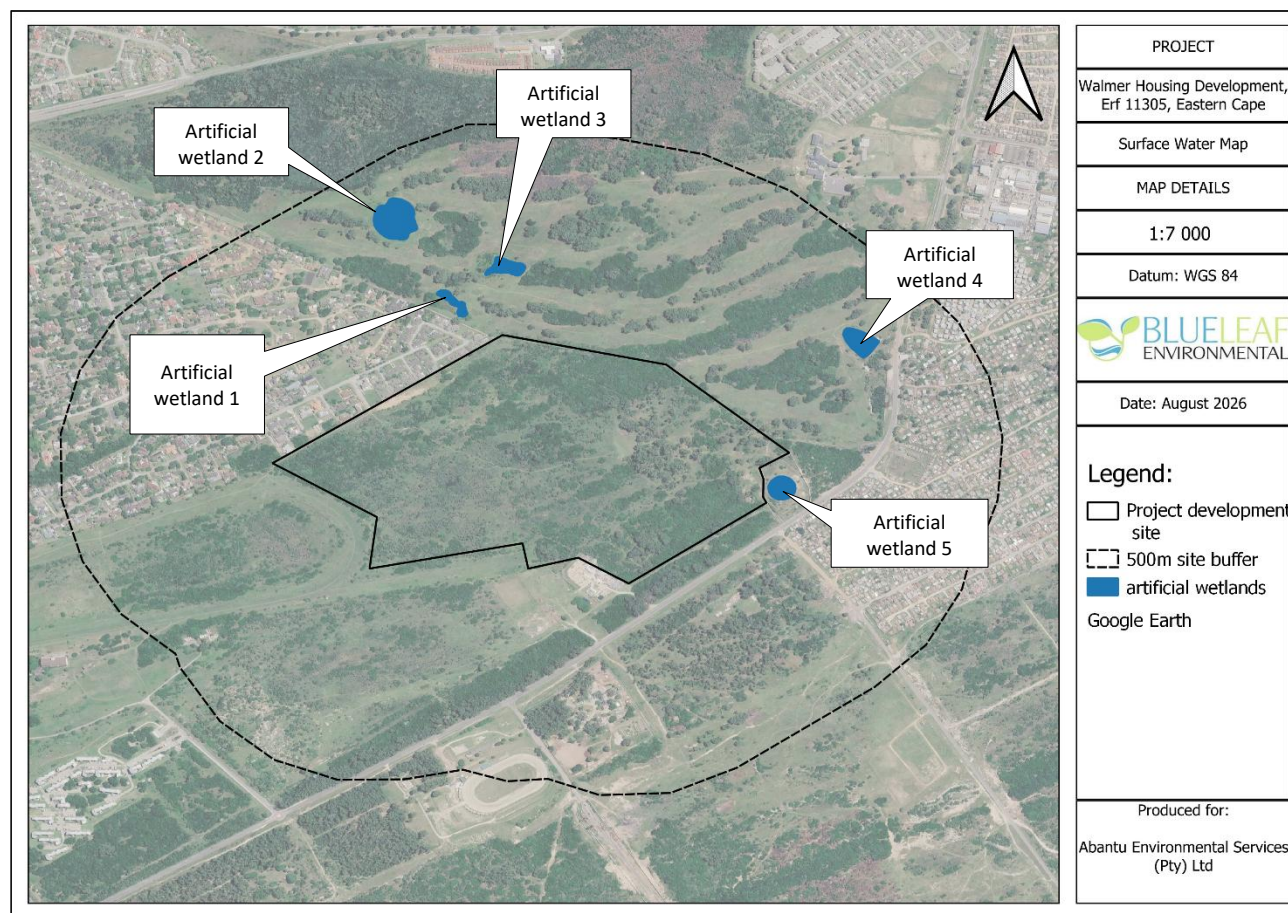


Figure 3.8: Delineated surface freshwater map of the PAOI

According to the surface water map (Fig 3.8), five artificial wetlands occur within the Total PAOI. A desktop analysis as well as a site inspection has confirmed the presence of these artificial wetland systems.

All of the wetland systems are classified and identified as artificial wetlands with artificial wetland 5 classified as a reservoir as per site inspection and desktop analysis. Artificial wetlands 1 – 4 are scattered across a green landscape which appears to be a golf course. These are man-made water features (such as golf course hazards, retention dams, or specialized stormwater management basins). In South African urban environmental planning, they serve crucial ecological functions by regulating urban stormwater run-off, filtering pollutants from nearby formal and informal settlements, and preventing downstream flooding.

Because these wetlands are situated close to dense urban settlements and disturbed soil zones, they are highly prone to colonization by invasive species such as *Acacia cyclops* (Rooikrans) and *Acacia saligna* (Port Jackson), which compete with the indigenous fynbos and alter local water availability.

Since these wetland systems have been identified as artificial, they will be excluded from the wetland delineation process. No rivers were observed within the Total PAOI.

4. Freshwater classification

South Africa is a geologically, geomorphologically, climatically, and ecologically complex country, and this has resulted in a diverse range of ecosystems, including rivers. The area investigated included the study site and a 500 m buffer area around the study site boundary. Various dams and rivers were delineated within the PAOI (Figure 5.1).

Each freshwater system was classified as per the DWS (2005) guidelines for “*a practical field procedure for delineation of wetlands and riparian areas*” as recommended in Gazette No. 19182, Notice No. 1091 of the National Water Act, 1998.

4.1. Level 1: System setting

Aquatic features within the PAOI are all classified as Inland Systems. An Inland System is defined as an aquatic ecosystem with no existing connection to the ocean. These ecosystems are characterized by the complete absence of marine exchange and/or tidal influence.

Two types of Inland Systems exist:

1. **Rivers**, which are ‘lotic’ aquatic ecosystems with flowing water concentrated within a distinct channel, either permanently or periodically. No rivers occur within the Total PAOI.
2. **Wetlands**, which are transitional between aquatic and terrestrial systems, and are generally characterized by (permanently to temporarily) saturated soils and hydrophytic vegetation. These areas are, in some cases, periodically covered by shallow water and/or may lack vegetation. No wetlands occur within the Total PAOI
3. **Open waterbodies**: Five artificial wetlands/dams were located within the PAOI (Fig 3.6). These artificial dams are described as open waterbodies which are permanently inundated ‘lentic’ aquatic ecosystems where standing water is the principal medium within which the dominant biota live.

4.2. Level 2: Regional setting

Various spatial frameworks can be used to categorize the regional setting of an Inland System. Both the DWS Ecoregions as well as the NFEPA WetVeg groups will be used in this report. Ecoregions describe aquatic systems in the context of national water resource management while the NFEPA WetVeg, as a provincial vegetation map, are appropriate for fine-scale wetland conservation planning.

Ecoregions

River ecoregional classification or typing allows the grouping of rivers according to similarities based on a top-down nested hierarchy. The principle of river typing is that rivers grouped together at a particular level of the typing hierarchy will be more like one another than rivers in other groups. Ecological regions are regions within which there is relative similarity in the mosaic of ecosystems and ecosystem components (biotic and abiotic, aquatic, and terrestrial). According to the Department of Water and Sanitation Level 1 River Ecoregional Classification System (DWS; 2005), the study area falls

in within **Ecoregion 20: South-Eastern Coastal Belt**. This region has a diverse topography. Although plains occur in the region, closed hills and mountains primarily characterize the topography with a moderate to high relief.

Major rivers within the ecoregion include the Swartkops, Gamtoos and Keurboos Rivers, but none of these are located nearby the site.

This ecoregion has the following characteristics:

Mean annual precipitation: Moderate to high.

Coefficient of variation of annual precipitation: Low to moderate

Drainage density: Low to medium

Stream frequency: Low/medium to medium/high in limited areas

Slopes: <5%: >80% but significant areas <20%

Median annual simulated runoff: Moderate to very high.

Mean annual temperature: Moderate to moderately hot.

Quaternary Catchment

The project area is located within the M20A quaternary catchment. The main aquatic feature within this catchment is the artificial wetlands within the Total PAOI, the site falling within a SWSA (Strategic Water Source Area due to the Baakens river flowing north outside the Total PAOI) as well as the site falling with an ESA1.

This quaternary catchment forms part of the Mzimvubu Tsitsikamma Water Management Area (WMA2023) that is important as a freshwater ecosystem priority area. Major rivers include the Mzimvubu, Mtata, Mbashe, Buffalo, Nahoon, Groot Kei and Keiskamma, Fish, Kowie, Boesmans, Sundays, Gamtoos, Kromme, Groot and Tsitsikamma.

NFEPA WetVeg

The Vegetation Map of South Africa, Swaziland and Lesotho (Mucina & Rutherford; 2018) groups vegetation types across the country according to Biomes (for example, Grassland or Savanna), which are then divided into Bioregions. To categorise the regional setting for the wetland component of the NFEPA project, wetland vegetation groups (referred to as WetVeg Groups) were derived by further splitting Bioregions into smaller groups through expert input. There are currently 133 NFEPA WetVeg Groups in South Africa.

The Total PAOI is located within one wet vegetation type namely: Eastern Fynbos Renosterveld Sandstone Fynbos which is classified as **Critically Endangered**. This is the threat classification for the site according to the NEMBA Revised National List of Ecosystems that are Threatened and in need of Protection (G. NR. 2747, 18 November 2022), which is the latest gazetted list of threatened ecosystems available for the country. It is based on the VegMap (2018) vegetation classification, published under Dayaram et al. (2019).

Wetlands

The site has a NWM5 Wetland Confidence of 4, indicating a medium to high level of confidence on the accuracy that the extent and HGM unit are well represented. A site visit confirmed no natural wetlands present on site with five artificial wetlands/dams occurring within the PAOI.

Rivers

No rivers occur within the Total PAOI.

4.3 Level 3: Landscape setting

Landscape settings are included because hydrological and hydrodynamic processes acting within a specific Inland System are strongly influenced by their topographical positions within the landscape, and by the geomorphological processes that have brought about and drive those topographical contexts. The following regional landscape units were determined for each of the freshwater features found within the PAOI.

Water feature (as in Figure 3.10)	Level 3: Landscape setting	Description of landscape setting
Artificial wetlands 1 - 5	Plain	Plain – an extensive area of low relief. These areas are generally characterised by relatively level, gently undulating or uniformly sloping land with a very gentle gradient that is not located within a valley. Gradient is typically less than 0.01 or 1:100.

4.4. Level 4: HGM Unit

Level 4 of the Classification System allows you to identify the Hydrogeomorphic (HGM) Units within an inland aquatic ecosystem. HGM Units are distinguished primarily based on:

1. Landform, which defines the shape and localised setting of the aquatic ecosystem.
2. Hydrological characteristics, which describe the nature of water movement into, through and out of the aquatic ecosystem.
3. Hydrodynamics, which describe the direction and strength of flow through the aquatic ecosystem.

Of the seven primary HGM Types the following are recognized on site and are categorized as follows:

System name	HGM TYPE	Longitudinal Zonation/Landform/ Outflow Drainage	Landform/Inflow Drainage
	Level 4A	Level 4B	Level 4C
N/A	N/A	N/A	N/A

4.5. Level 5: Hydrological regime

While the HGM Unit is influenced by the source of water and how it moves into, through and out of an Inland System, the hydrological regime describes the behavior of that water within the system and, for wetlands, in the underlying soil.

River flow-types (flow regimes)	Description	
	Level 5A	Level 5B
Perennial	Flows continuously throughout the year, in most years. No perennial rivers occur within the project development site.	N/A

River flow-types (flow regimes)	Description	
	Level 5A	Level 5B
Non-Perennial	Does not flow continuously throughout the year, although pools may persist. Flow regime is unpredictable. No non-perennial rivers are located within the project development site.	N/A

Hydroperiod categories (for non-river inland systems)

For all Inland Systems that are not rivers (i.e. wetlands and open waterbodies), the hydrological regime is classified according to the period of inundation (at Level 5A), and saturation (At Level 5B), together with the inundation depth class (at Level 5C) in the case of permanently inundated open waterbodies. (Table 5.2).

Table 5.2: Hydroperiod categories for non-river inland systems

Level 5: Hydroperiod and depth of inundation			
	Level 5A	Level 5B	Level 5C
	Inundation periodicity	Saturation periodicity (within 0.5m of soil surface)	Inundation-depth class
Artificial wetlands 1 - 5	Seasonally inundated	Seasonally saturated	Unknown

4.6. Level 6: Descriptors

Descriptors describe the structural/chemical/biological characteristics. The following table defines the relevant descriptors within the Total PAOI:

Descriptors	Description		
Natural vs Artificial	System name	Classification	Artificial subcategories
	Artificial wetland 1 to 5	Artificial wetlands	N/A
Salinity	Salinity category	Salinity (or TDS) range	Conductivity range
	Fresh for all systems	Not measured. Estimate <3.0 g/l	Not measured, Estimate <500 mS/m
pH	Action taken	pH category	pH range
	Not measured	Anticipating circum-neutral for all	pH of 6-8
Substratum type	Level 6A	Level 6B	Comments
	Sandy soil	N/A	
Vegetation	Vegetation cover	Vegetation form	Vegetation description
	Vegetated for most artificial systems	Herbaceous	Reeds
Geology	Refer to section 3.3		

4.6. Ecological Importance and Sensitivity (EIS)

While the wetland features are artificial, they may possibly fulfill an important ecological function, which should be assessed prior to developing the site. The Ecological Importance and Sensitivity (EIS – Rountree and Kotze, 2013) was therefore used to assess the artificial wetlands in terms of their ecological and hydro-functional importance and their ability to provide direct human benefits (see the Appendix for the methodological approach).

Table 1. Ecological Importance and Sensitivity importance criteria for the artificial wetlands observed on the residential development, St Francis, Eastern Cape.

Determinant	Score	Confidence	Comments
1. Rare & Endangered Species	0	2	Low probability due to wetlands being small and artificial.
2. Populations of Unique Species	1	3	Large populations of unique wetland species are very small given small size (although 1 large artificial wetland is present) and temporary hydroperiod.
3. Species/taxon Richness	1	3	Small diversity of bird, mammal and insect species present
4. Diversity of Habitat Types or Features	1	3	Small or low diversity due to vegetation being transformed with some vegetation maintained as short grasses.
5 Migration route/breeding and feeding site for wetland species	1	4	Localized diversity of bird species.
6. Sensitivity to Changes in the Natural Hydrological Regime	0	5	Artificial wetlands not sensitive to changes in floods or low flows
7. Sensitivity to Water Quality Changes	1	5	Low sensitivity to changes in water quality
8. Flood Storage, Energy Dissipation & Particulate/Element Removal	1	5	Limited attenuation of stormwater runoff
9. Protected Status	2	5	Not formally protected but does however occur within a ESA1.
10. Ecological Integrity	0	4	Artificial wetlands
Total	0.8	3.9	
Median	1	4	
Overall ecological sensitivity & importance	1	4	Low/Marginal: Not ecologically important and sensitive at any scale. Biodiversity is ubiquitous and not sensitive to flow and habitat modifications. (Wetlands: play an insignificant role in moderating water quality and quantity.

The Ecological Importance and Sensitivity (EIS) is an expression of its importance to the maintenance of ecological diversity and functioning on local and wider scales, and both abiotic and biotic components of the riverine and wetland system are taken into consideration. Sensitivity refers to the system's ability to resist disturbance and its capability to recover from disturbance once it has occurred. **The artificial wetlands found within the PAOI are all categorized as Low**, which is not ecologically important and sensitive at any scale. Biodiversity is ubiquitous and not sensitive to flow and habitat modifications. (Wetlands: play an insignificant role in moderating water quality and quantity.

Given the relatively small size of the artificial wetlands and their temporary hydroperiod they are unlikely to be of significant ecological importance and provide limited habitat for aquatic biota (Table 1). Due to their artificial and temporary, endhoreic nature, the wetlands are not sensitive to changes in floods or flow conditions and are marginally sensitive to changes in water quality. Their temporary hydroperiod, endorheic flow pattern and isolation from a larger hydrological network provides no streamflow regulation function (e.g. sustaining flow during low flow periods) but does provide limited capacity for flood attenuation and assimilation of pollutants and nutrients derived from stormwater runoff. Finally, their small size, temporary hydroperiod and low density of wetland plant species provides limited direct human benefits.

4.9. Present Ecological State (PES)

An important factor that influences the diversity and abundance of aquatic communities is the condition of the surrounding physico-chemical habitat. Habitat loss, alteration, or degradation generally results in a decline in species diversity and the PES is determined by assessing the extent to which habitat has degraded relative to the natural reference condition. The PES can therefore only be determined for natural systems. The PES cannot be determined for artificial systems as there is no reference condition against which to assess changes in habitat over time.

5. Riverine and Wetland Delineation

Since no rivers or natural wetlands were present on site or within 100m or 500m of the site respectively, no riverine or wetland delineations were required. All artificial wetlands were delineated and are shown in Figure 6.1 below.

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6. Site Ecological Importance

Site Ecological Importance (SEI) was determined for the proposed construction of Walmer Housing Development on Erf 11305, Port Elizabeth, in the Eastern Cape Province. Environmental constraints were identified and aligned with specific aquatic characteristics of the site. The following site characteristics contributed to determining an overall sensitivity:

Site characteristic	Description of characteristic	Sensitivity allocation	Feature
River systems	No perennial or non-perennial river systems were identified	High	All rivers and drainage systems are classified as highly sensitive. The risk of impacts, indirect and cumulative impacts, are high.
	100m River Regulated Areas	Low	A 100m buffer around all rivers is considered low which means there is a low risk of indirectly impacting the rivers.
Wetland systems	Wetlands (x 0)	High	All Wetland features are protected and sensitive in South Africa as it presents a high risk of impacts.
	500m Wetland Regulated Areas	Low	A 500m buffer around all wetlands is considered low which means there is a low risk of indirectly impacting the wetlands.
Vegetation	Riparian zone	High	Areas consisting of dense riparian vegetation is considered as high site ecological importance.

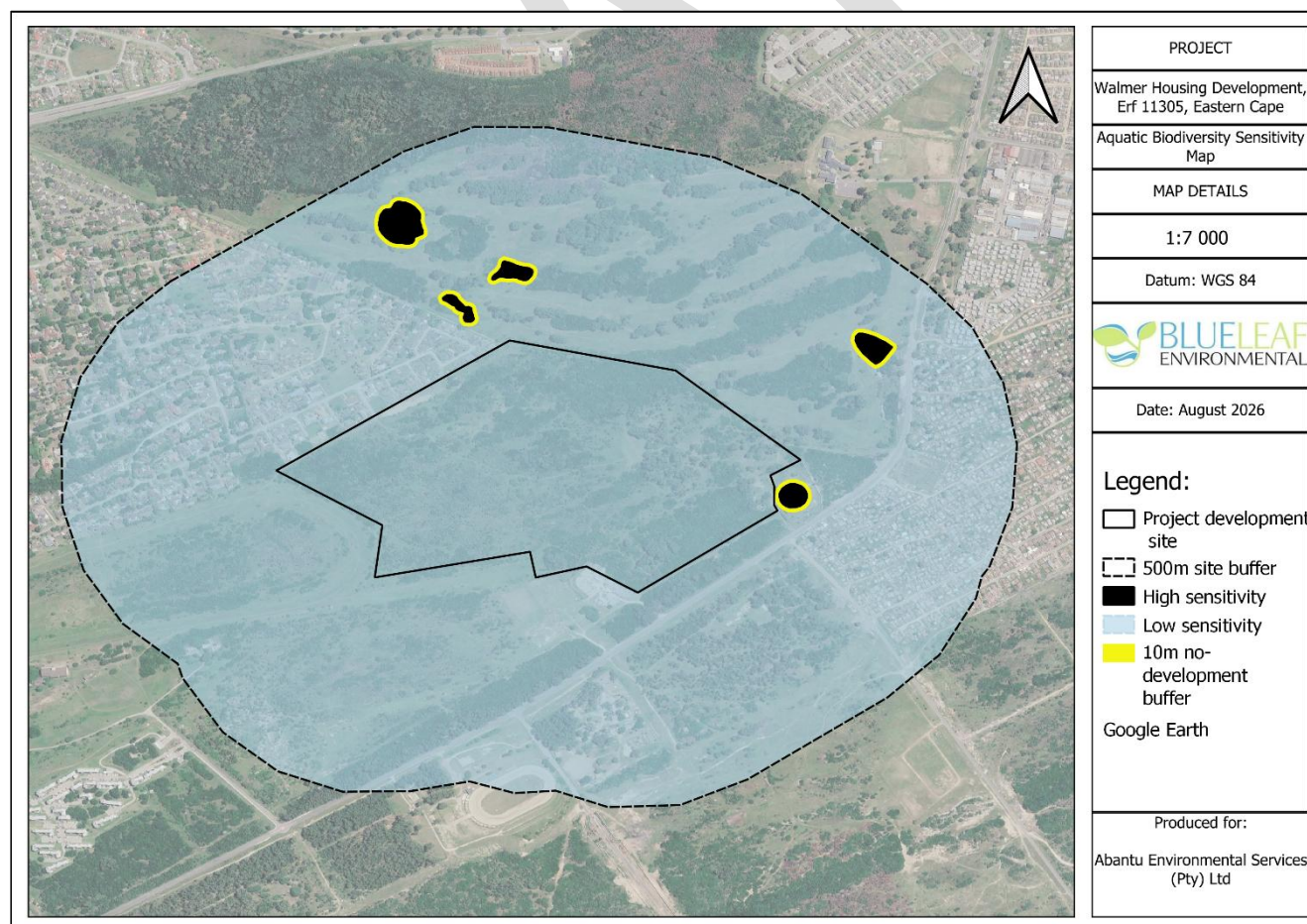


Figure 6.1: Sensitivity Map for the proposed development area and surroundings.

A 10m no-development buffer was applied to watercourses potentially affected by the development. These buffer zones have been defined as a strip of land with a use, function or zoning specifically designed to act as barriers between human activities and sensitive water resources with the aim of protecting these water resources from adverse negative impacts. Appropriate buffers were estimated based on buffer zone guidelines developed by Macfarlane and Bredin (2017). These guidelines estimate required buffer zone widths based on a combination of input parameters which include, *inter alia*, the nature of the activity and associated impacts, basic climatic and soil conditions and the implementation of appropriate mitigation measures.

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7. Impact Assessment

The following issues were identified during the aquatic biodiversity assessment of the proposed construction of Walmer Housing Development on Erf 11305, Port Elizabeth, in the Eastern Cape Province.

7.1 Identified impacts.

The following aquatic issues were identified during the assessment of the development area:

#	Activity causing impact (Issue)	Description of impact
1.	Non-compliance with existing legislation	1.1. Legal compliance Non-compliance with ecological laws and policies of South Africa could lead to unnecessary delays in establishment activities, and potentially criminal cases, based on the severity of the non-compliance, being brought against the proponent and his/her contractors.
2.	Operation of machinery and vehicles	2.1. Hydro chemical pollution of surface waterbodies Hydrochemical spills from machines and vehicles may contaminate water quality of the surface waterbodies within the Total PAOI.
		2.2. Chemical pollution of surface waterbodies Wet concrete spillage from infrastructure and internal roads may lead to the contamination (impact on surface water quality) of the surface waterbodies within the Total PAOI.
3.	Construction and clearing of vegetation	3.1. Loss of vegetation Construction and clearing of vegetation may lead to the regrowth of alien vegetation impacting ecological functioning of the surface waterbodies within the Total PAOI.
		3.2. Increased risk of soil erosion (loss of stabilizing vegetation) Clearing vegetation leads to the increased risk of soil erosion which may impact on the ecological functioning of the surface waterbodies within the Total PAOI.
4.	Loss of biodiversity	4.1. Contamination of the local stormwater system Contaminated stormwater from the project site may contaminate the surrounding freshwater system.
5.	Soil stockpiling	5.1. Sedimentation of surface waterbodies Soil stockpiles within or near the artificial wetland systems leads to increase in sediment inputs and turbidity, alteration to flow volumes and alteration of patterns of flow (increased flood peaks).

All impacts identified above were assessed as per the assessment methodology described in Chapter 2.8 of this report. Each impact was described on how it will impact within a specific phase of the project, namely Planning and Design, Construction and Operation.

Issue 1:	Non-compliance with existing legislation
Consequence of Issue	Non-compliance with aquatic laws and policies of South Africa could lead to unnecessary delays in establishment activities, and potentially criminal cases, based on the severity of the non-compliance, being brought against the proponent and his/her contractors. Permits will be required for all infrastructure within 500 m of a wetland and within 100 m of a river. Further permits will be required if any development footprint is to be located within a wetland system.
Number of impacts identified associated with this issue	Only 1 (Impact 1.1 only)

Impact 1.1: Legal compliance			
Phase of expansion: Planning and Design Phase			
Nature of impact	Non-compliance with aquatic laws and policies of South Africa could lead to unnecessary delays in establishment activities, and potentially criminal cases, based on the severity of the non-compliance, being brought against the proponent and his/her contractors.		
Cumulative impact	None		
Indirect impacts	None		
Residual impacts	None		
Classification of impact	Before mitigating	After mitigating	Consequence of Impact
Duration of impact	5	2	During all phases.
Extent of impact	5	1	DWS approval will be required.
Intensity of impact	5	2	Legislated approval is required for all infrastructure within 500 m of a wetland and within 100 m of a drainage system.
Severity	15	5	Duration + extent + intensity
Probability of impact occurring	5	2	Impact will occur on commencement of construction.
Frequency	5	1	Impact will occur once only.
Incidence	10	3	Frequency + Probability
Degree of reversibility	High		Permits will be required.
Irreplaceability	Low		No resource will be lost.
Mitigations	Mitigatory potential		Recommended mitigations
	High		- Ensure compliance with all relevant environmental regulations and permits. - All relevant DWS permits (GA and/or full WULA) must be obtained from the competent authorities prior to commencement of any activity on site. - Any footprint within 500m of a delineated wetland will require a GA application with DWS. - Any footprint within 100m of a delineated river will require a GA application with DWS. - Any footprint within a delineated wetland or river will require a full WULA application with DWS.
Significance of impact (Severity x Incidence)	Pre-mitigation significance		Post-mitigation significance
	Very high negative (150)		Very low negative (15)

Issue 2:	Operation of machinery and vehicles
Consequence of issue	Hydro-chemical and chemical spills from machines and vehicles during construction may leak (runoff) into the surrounding artificial wetlands contaminating water quality.
Number of impacts	2 (Impact 3.1 and 3.2)

Impact 2.1: Hydro-chemical pollution of surface waterbodies			
Phase of expansion: Construction Phase			
Nature of impact	Leaks (oil, diesel, and petrol) from vehicles working or parked may contaminate water quality of the artificial wetlands within the Secondary PAOI.		
Cumulative impact	Increasing levels of contamination of the artificial wetlands within the Secondary PAOI.		
Indirect impacts	Soil contamination.		
Residual impacts	Pollution of waterbodies.		
Classification of impact	Before mitigating	After mitigating	Consequence of Impact
Duration of impact	3	2	Short term impact.
Extent of impact	1	1	Limited to project site and boundary of site.
Intensity of impact	2	2	It can lead to loss of vegetation.
Severity	6	5	Duration + extent + intensity
Probability of impact occurring	5	2	Impact would occur at construction phase
Frequency	5	1	
Incidence	10	3	Frequency + Probability
Degree of reversibility	Low negative		Impacts can be easily reversed.
Irreplaceability	Medium		
Mitigations	Mitigatory potential		Recommended mitigations
	High		- Do not park any construction machinery overnight close to any surface waterbodies. - Parked vehicles must always have drip trays underneath it. These drip trays should be cleaned regularly, and the waste removed from site. - Stored hydro-chemicals must be kept off site or in bunds away from any surface waterbody. - All waste products must be stored at least 5m away from any surface waterbody and regularly removed from site to a registered landfill by a qualified contractor.
Significance of impact (Severity x Incidence)	Pre-mitigation significance		Post-mitigation significance
	Medium high (60)		Very Low negative (15)

Impact 2.2: Chemical pollution of surface waterbodies			
Phase of expansion: Construction Phase			
Nature of impact	Wet concrete spillage may lead to the contamination of the artificial wetlands within the Secondary PAOI.		
Cumulative impact	None		
Indirect impacts	Soil contamination.		
Residual impacts	Pollution of waterbodies.		
Classification of impact	Before mitigating	After mitigating	Consequence of Impact
Duration of impact	3	2	It would be ongoing during operations.
Extent of impact	2	1	Limited to project site and boundary of site.
Intensity of impact	3	2	It can lead to loss of vegetation.
Severity	8	5	Duration + extent + intensity
Probability of impact occurring	4	2	Impact may possibly occur on commencement of construction.
Frequency	5	1	Daily risk of impact occurring.
Incidence	9	3	Frequency + Probability
Degree of reversibility	Low negative		Impacts can be easily reversed.
Irreplaceability	Medium		Impact will contribute to existing contamination levels.
Mitigations	Mitigatory potential		Recommended mitigations
	High		- Do not mix any wet concrete on site. Concrete mixing should be done offsite and then transported to site. - If not possible, then do not mix any wet concrete at least 5m away from any surface waterbody. - Apply methods with a low risk of spillage and leaking during concrete pours. - Do not mix on bare ground. All mixing must take place on a hard surface.
Significance of impact (Severity x Incidence)	Pre-mitigation significance		Post-mitigation significance
	Medium high (72)		Very low negative (15)

Issue 3:	Construction and clearing of vegetation
Consequence of issue	Clearing of vegetation leads to increase in alien vegetation and erosion.
Number of impacts	2 (Impact 3.1 and 3.2)

Impact 3.1: Loss of vegetation			
Phase of expansion: Construction Phase			
Nature of impact	Construction and clearing of vegetation may lead to the regrowth of alien vegetation impacting ecological functioning of the artificial wetlands within the Secondary PAOI.		
Cumulative impact	Loss of vegetation.		
Indirect impacts	Loss of habitats and connectivity and regrowth of alien vegetation.		
Residual impacts	Loss of ecological functioning of the artificial wetlands within the Secondary PAOI.		
Classification of impact	Before mitigating	After mitigating	Consequence of Impact
Duration of impact	3	1	Clearing of vegetation only during construction phase
Extent of impact	4	1	Local area where construction is to take place.
Intensity of impact	4	2	Leading to permanent loss of vegetation
Severity	11	4	Duration + extent + intensity
Probability of impact occurring	5	1	Impact will occur on commencement of construction.
Frequency	5	1	Daily risk of impact occurring (only during construction phase).
Incidence	10	2	Frequency + Probability
Degree of reversibility	High negative		Impacts can be reversed easily.
Irreplaceability	Medium		
Mitigations	Mitigatory potential		Recommended mitigations
	High		– An Alien Vegetation Management Plan must be developed for all phases of development. – Remove alien vegetation on site. this must be done during all phases of development.
Significance of impact (Severity x Incidence)	Pre-mitigation significance		Post-mitigation significance
	Medium High (110)		Very low negative (8)

Impact 3.2: Increased risk of soil erosion			
Phase of expansion: Construction Phase			
Nature of impact	Clearing of vegetation leads to the increased risk of soil erosion which may impact on the ecological functioning of the artificial wetlands within the Secondary PAOI.		
Cumulative impact	Increased risk of soil erosion.		
Indirect impacts	Sedimentation of rivers.		
Residual impacts	Soil and bank erosion.		
Classification of impact	Before mitigating	After mitigating	Consequence of Impact
Duration of impact	5	2	Duration of cleared vegetation.
Extent of impact	3	2	Limited to project site and boundary of site.
Intensity of impact	4	2	Risk increases continuously when vegetation is cleared.
Severity	12	6	Duration + extent + intensity
Probability of impact occurring	5	3	Impact will occur on commencement of construction.
Frequency	2	1	Impact will only occur during the construction phase
Incidence	7	4	Frequency + Probability
Degree of reversibility	Low negative		Impacts can easily be reversed.
Irreplaceability	Medium		
Mitigations	Mitigatory potential		Recommended mitigations
	High		- Develop and implement an Erosion Management Plan for both construction and operational phases. - Revegetate cleared areas as soon as possible.
Significance of impact (Severity x Incidence)	Pre-mitigation significance		Post-mitigation significance
	Medium high (84)		Very low negative (24)

Issue 4:	Loss of Biodiversity
Consequence of issue	Contaminated stormwater, especially after rain events from the project site will contaminate the surrounding freshwater system.
Number of impacts	1 (Impact 4.1 only)

Impact 4.1: Contamination of the local stormwater			
Phase of expansion: Construction and Operational Phase			
Nature of impact	Operations of the proposed development will result in the contamination of stormwater		
Cumulative impact	Contamination of the surrounding freshwater systems		
Indirect impacts	Contamination of the surrounding freshwater systems		
Residual impacts	Contamination of stormwater, especially after rain events		
Classification of impact	Before mitigating	After mitigating	Consequence of Impact
Duration of impact	5	3	Impact will be permanent.
Extent of impact	5	3	Site and surrounding areas will be impacted.
Intensity of impact	5	3	Impact on ecological functions.
Severity	15	9	Duration + extent + intensity
Probability of impact occurring	5	3	Impact will occur
Frequency	5	3	Throughout construction (continuously)
Incidence	10	6	Frequency + Probability
Degree of reversibility	High		Impact can be avoided by implementing the proposed mitigation measures.
Irreplaceability	Moderate		Functionality changes will be temporary.
Mitigations	Mitigatory potential		Recommended mitigations
	Medium		- Where feasible, the river and wetlands within the Secondary PAOI should be protected from polluted runoff by cut-off drains and/or similar interventions. - Residential stormwater runoff that exceeds the 1:5-year flood volumes can be conveyed directly into drainage, if of acceptable quality. All other stormwater runoff, including runoff from the first seasonal rains, should be directed elsewhere or treated onsite, such that the water quality has been polished to an acceptable standard.
Significance of impact (Severity x Incidence)	Pre-mitigation significance		Post-mitigation significance
	Very High Negative (150)		Medium negative (54)

Issue 5:	Soil stockpiling
Consequence of issue	Soil stockpiles close to or within the artificial wetland systems can lead to increased sediment input and turbidity.
Number of impacts	1 (Impact 5.1 only)

Impact 5.1: Sedimentation of surface waterbodies			
Phase of expansion: Construction Phase			
Nature of impact	Placing soil stockpiles close to or within the artificial wetland systems can lead to increased sediment inputs and turbidity.		
Cumulative impact	Increased sediment input		
Indirect impacts	Turbidity/loss of vegetation		
Residual impacts	Stockpiles washing into nearby artificial wetlands/nearby rivers		
Classification of impact	Before mitigating	After mitigating	Consequence of Impact
Duration of impact	4	1	Impact will be for a short time and only happen during the construction phase.
Extent of impact	2	1	Impact limited to study area.
Intensity of impact	5	1	Fluvial processes may be altered
Severity	15	9	Duration + extent + intensity
Probability of impact occurring	5	2	Will definitely happen if stockpiled in the artificial wetland system
Frequency	5	3	Will occur during construction phase
Incidence	10	6	Frequency + Probability
Degree of reversibility	High		Impact could be avoided by implementing the proposed mitigation measures.
Irreplaceability	Moderate		River and wetland systems are sensitive to fluvial changes.
Mitigations	Mitigatory potential		Recommended mitigations
	Medium		– Do not place any stockpiles within or near the wetland systems to prevent contamination/sedimentation of the wetland systems.
Significance of impact (Severity x Incidence)	Pre-mitigation significance		Post-mitigation significance
	Very High Negative (110)		Very low negative (20)

8. Conclusion

BlueLeaf has been appointed by Abantu to conduct an Aquatic Biodiversity and Delineation Report as part of the EIA process conducted by Abantu. This report also complies with the DWS requirements for a wetland delineation and assessment report to confirm the presence of wet conditions and the extent thereof on the proposed development.

This investigation has been undertaken to form part of the NEMA as amended, as well as the IWULA to be submitted to DWS. The establishment of any type of development on the extent of the rivers, wetland systems and floodplains within the property poses detrimental environmental impacts based on the nature and magnitude of the development and legal requirements in the form of the NEMA, and the NWA.

The proposed housing development will be located on Erf 11305, Port Elizabeth, in the Nelson Mandela Bay Metropolitan Municipality (Port Elizabeth/Gqeberha) in the Eastern Cape Province.

The development site is located between: the suburbs of Walmer Heights and Beaumont Estates, to the west and northwest; the old Arlington Racecourse, to the west; SmartStone Port Elizabeth, to the south; the Walmer Country Club, to the north and northeast; Walmer Township, to the east, and Victoria Drive (M18) and open space, to the southeast.

The project will involve construction of the Walmer Housing Development, including associated facilities and infrastructure. The project site is 43.73 hectares in total extent.

The Gross Residential Area (single dwellings) is 11.11 hectares – with a Gross Residential Density of 14.9 units per hectare (single dwellings) and a Nett Residential Density of 58.69 units per hectare (single dwellings). The average erf size for single dwellings is 170.41 m². No internal service infrastructure is currently in place on site – however, electrical, sewage and water infrastructure will be connected to the existing bulk infrastructure currently servicing the surrounding residential areas.

The housing development will provide a mix of housing typologies and associated facilities which include the following:

Housing typologies and associated facilities:

- Free Basic House/RDP
- GAP/FLISP
- Social Housing
- Open/GAP market housing

Other land uses:

- Business Zone I: to provide for businesses such as a supermarket, bottle store, service trades etc. with parking provision at 3.33 bays per 100 m²
- Institutional Zone I: school site
- Authority Zone: authority uses such as police station, fire services, telecommunications facilities etc.
- Open Space Zone I: public open space.
- Special Use: conservation usage (forest patches)
- Transportation Zone II: public zone.

The total area of the site is 43.73 ha. The Gross Residential Area (single dwellings) is 11.11 ha with Gross Residential Density (single dwellings) being 14.90 units/ha and a Nett Residential Density (single dwellings) being 58.69 units/ha. The average erf size for single dwellings is 170.41m². No internal infrastructure is currently in place on site; however electricity, sewer and water will be connected onto existing bulk infrastructure currently servicing the surrounding areas.

The Project Areas of Influence (PAOI) has been determined as follows:

PAOI	Area (ha)	Description	Probability of impact occurring
Primary PAOI	43.6 ha	The Primary PAOI includes all areas within the boundaries of the project development site. This is the area directly impacted by the proposed development and includes both permanent and temporary footprints.	Definite
Secondary PAOI	225.4 ha	The secondary PAOI includes all areas within a 500 m buffer of the proposed development boundary. These areas are not directly impacted by the development unless temporary footprints like site camps, laydown areas and stockpiles are placed in them. Assessing this PAOI will not only result in identifying potential indirect and cumulative impacts but will also allow for micro-movement of infrastructure.	Unlikely
Total PAOI	269 ha	The Primary and Secondary PAOI's are collectively referred to as the Total PAOI (or just the PAOI) or Study Site in this report and demarcate the extent of the study site that will be assessed.	Likely

The freshwater features within the Total PAOI are summarised below:

Descriptors	Description		
Natural vs Artificial	System name	Classification	Artificial subcategories
	Artificial dam 1 to 5	Artificial wetlands	N/A
Salinity	Salinity category	Salinity (or TDS) range	Conductivity range
	Fresh for all systems	Not measured. Estimate <3.0 g/l	Not measured, Estimate <500 mS/m
pH	Action taken	pH category	pH range
	Not measured	Anticipating circum-neutral for all	pH of 6-8
Substratum type	Level 6A	Level 6B	Comments
	Sandy soil	N/A	
Vegetation	Vegetation cover	Vegetation form	Vegetation description
	Vegetated for most artificial systems	Herbaceous	Reeds
Geology	Refer to section 3.3		

This section only addresses aquatic related biodiversity sensitive areas. Terrestrial biodiversity sensitive area is addressed in the Terrestrial Biodiversity Assessment for the project.

According to the ECBCP, north of the proposed development site occurs within an ESA1.

The sensitivity allocation is due to the ESA1 functioning as vital ecological infrastructure under the Eastern Cape Biodiversity Conservation Plan. The zone directly north of Erf 11305 contains the Baakens River valley system and its local drainage paths. In a freshwater conservation framework, these areas are flagged to shield water source areas, manage stormwater runoff, and control soil erosion from urban encroachment.

ESA's are important in supporting the functioning of CBAs and are often vital for delivering ecosystem services. ESA 1 are areas in a natural condition that are required to meet biodiversity targets, for species, ecosystems or ecological processes and infrastructure. ESA 2 are areas that are not essential for meeting biodiversity targets but are targeted for restoration and play an important role in supporting the functioning of Protected Areas (PAs) or CBAs and are often vital for delivering ecosystem services.

The site does not fall within any CBA2.

8.1. Site Ecological Importance

Site Ecological Importance of sections within the study site allocated a high sensitivity to all water related structures (rivers, wetlands and riparian zones) within the PAOI. No development is allowed in these areas without prior approval from DWS. The remainder of the PAOI has a low sensitivity. Any activity occurring within 100m of a river or 500m of a wetland will require a water use application with DWS. A 10m no development buffer has been placed around all artificial wetlands within the Total PAOI as all water resources within South Africa is recognized as sensitive.

8.2. Alternatives

No site alternatives or layouts are proposed.

8.3. Cumulative impacts

In terms of Environmental Impact Assessment, Cumulative Impact is defined as:

"Means the past, current and reasonably foreseeable future impact of an activity, considered together with the impact of activities associated with that activity, that in itself may not be significant, but may become significant when added to the existing and reasonably foreseeable impacts eventuating from similar or diverse activities".

The following cumulative impacts were identified:

1. Loss of freshwater.
2. Increasing levels of contamination of artificial wetlands within the PAOI.

8.4. Levels of acceptable change

The proposed development is considered as an acceptable change to the environment provided all proposed mitigations are implemented.

8.5. Levels to be avoided.

All freshwater systems must be avoided. Provided that all mitigation measures proposed in this report are implemented, these risks will be reduced and therefore considered as an acceptable change to the local environment.

8.7. General rehabilitation measures

- Apply for a Section 21 c&i Water Use Application with DWS for developing within 500m of a wetland.
- Apply for a full WULA if developing inside an existing wetland or river.
- Stringent adherence to Environmental Management Plans (EMP's) for both construction and operational phases to minimize damage.
- Develop an Alien Invasive Management Plan to identify, control, and eradicate non-indigenous species to protect biodiversity and ecosystem functions.

9. Specialist Opinion

The proposed development is NOT considered to be fatally flawed, and no components of the project have been identified as flawed.

The combination of the desktop survey, analysis of historical imagery and site visit found no evidence of the presence of any natural watercourses on the property. All observed wetlands on site are artificial. While the ecological importance of these artificial wetlands is low, they are moderately important in terms of their hydro-functional attributes and could provide a useful function with respect to attenuating stormwater runoff.

However, no development must be allowed within the delineated artificial dams and the demarcated 10m no development buffer. These areas should be considered as No-Go Areas for construction activities. All construction proposed within these areas should be reconsidered to not infringe within these areas.

The proposed construction of Walmer Housing Development on Erf 11305, Port Elizabeth, in the Eastern Cape Province was assessed and considered to be acceptable, provided that all mitigation measures provided in this report are implemented. This includes all impacts that have already occurred, are currently occurring or may still occur.

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